Exact Compositional Transfer of Bounded Linear Recovery

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Abstract

We study which local recovery information suffices to compose bounded linear repairs through concatenated codes. The least helper count alone is insufficient: composition also requires the functional supplied by each local choice. We determine the exact minimum helper cost of a recovery system that leaves its target block, and prove an associative composition law that reconstructs coefficient witnesses from local choices. Equation escape need not create a new repair option, because external coefficients can be dispensable. If every outer dual word touching the target block has weight greater than $r+1$, all inclusion-minimal recovery supports of size at most r remain local, without an inner-dual-distance condition. This preserves the available bounded repair choices and transfers reliability under arbitrary availability laws with the same local marginal, as well as fractional and integral support allocations. Equation confinement and confinement of minimal repairs therefore have different requirements.

1 Introduction

A failed coded symbol may have several linear repairs. Choosing one with few helpers answers a local question; composing that repair with another code, or sharing helpers among requests, can require choices that the local minimum discarded. We ask which recovery data must be retained to answer these later questions exactly.

A running example. Let $u, v \in \mathbb{F}_2$ be independent message symbols. Store target u and four helpers

$$(h_1, h_2, h_3, h_4) = (u, u, v, u+v).$$

The minimal repairs of u are $\{1\}$, $\{2\}$, and $\{3,4\}$: without either copy of u, both remaining helpers are necessary. Locality records only the minimum size, one. Keeping both size-one repairs still loses the option needed when helpers 1 and 2 fail. If each request consumes one unit at every helper it contacts, unit capacities allow the three disjoint repairs to serve three requests at once; size-one repairs alone serve only two.

Composition asks another question: which message functional does a helper supply? Here h_1 supplies u, while h_3 supplies v. Both cost one, but a surrounding equation requiring v cannot substitute h_1. Functional labels preserve that compatibility; physical supports preserve failure and capacity choices; coefficients make the chosen repair executable. The example will recur at each of these changes of viewpoint.

Let $I \leq \mathbb{F}_q^E$ have a full-row-rank encoder $G = (G_P \mid G_J)$, where $E = P \sqcup J$ partitions targets and helpers, and assume $0 < \dim I < |E|$. Put $W_P = \operatorname{im} G_P \cap \operatorname{im} G_J$. A normalized recovery system for $0 \neq T \leq W_P$ consists of linear maps $\alpha : T \to \mathbb{F}_q^P$ and $\beta : T \to \mathbb{F}_q^J$ satisfying $G_P \alpha = \mathrm{id}_T$ and $G_J \beta = -\mathrm{id}_T$. Its helper cost is $|\operatorname{supp} \beta(T)|$. Thus $\dim T$ counts independent target combinations, which may be fewer than the erased positions.

Identify the message space with a finite extension $L/\mathbb{F}_q$ and fix its encoder $\iota : L \xrightarrow{\sim} I$. The *represented inner code* retains this identification. Its functional map is

$$\Phi_I : \mathbb{F}_q^E \longrightarrow L^*, \qquad \Phi_I(y)(a) = \langle y, \iota(a)\rangle, \quad L^* = \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q).$$

For a linear map $b : T \to L^*$, record the two fibre minima

$$\lambda_{I,T}(b) = \min_{\Phi_I Y = b} |\operatorname{supp} Y(T)|, \qquad \mu_{I,P,T}(b) = \min_{\substack{G_P\alpha = \mathrm{id}_T \\ \Phi_I(\alpha,\beta) = b}} |\operatorname{supp} \beta(T)|.$$

All minimizing maps are $\mathbb{F}_q$-linear, with domains $Y : T \to \mathbb{F}_q^E$, $\alpha : T \to \mathbb{F}_q^P$, and $\beta : T \to \mathbb{F}_q^J$. Only the displayed constraints are imposed in each minimum; empty minima are ∞. The target block pays only for helpers, whereas an external block pays for all coordinates it uses.

For an L-linear outer code $O \le L^N$, define

$$\mathrm{FD}(O) = \{(b_h) \in (L^*)^N : \textstyle\sum_h b_h(a_h) = 0 \text{ for every } a \in O\}.$$

These are the compatible block functionals. Trace duality identifies this space with $O^\perp$ and preserves block support. Write $O \circ I = \{(\iota(a_h))_{h=1}^N : a \in O\}$. A global normalized recovery system is a linear map $Y : T \to (O \circ I)^\perp$ whose target coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P\alpha = \mathrm{id}_T$. For a target block j, put

$$\Gamma_{j,T}(O, I) = \min\left\{ \mu_{I,P,T}(0) + d(I^\perp), \quad \min_{\substack{0 \ne B : T \to \mathrm{FD}(O) \\ B \text{ linear}}} \left[\mu_{I,P,T}(B_j) + \sum_{h \ne j} \lambda_{I,T}(B_h) \right] \right\},$$

and set $\Gamma_{j,t} = \min_{T \le W_P,\, \dim T = t} \Gamma_{j,T}$. We use $d(0) = \infty$.

Three information levels must be distinguished. Coefficient equations retain the supplied functionals and executable lifts; their support families retain repair alternatives and overlap; scalar minima retain only optimum values. Composition uses costs indexed by functional labels, while future failure and price queries require the support alternatives. The theorem below computes equation escape, including dispensable external coefficients.

Theorem 1.1 (Main characterization: exact finite transfer)**.** *Let $\ell = \dim W_P \ge 1$ and $1 \le t \le \ell$. Let I be used as the represented inner code in a concatenation with an L-linear outer code $O \le L^N$ with $N \ge 2$, where L is the message space of I, and fix a block j onto which O projects nontrivially. The minimum cost of a nonconfined recovery system—one using at least one helper outside block j—in recovered dimension t is $\Gamma_{j,t}(O, I)$. Hence confinement through radius r holds exactly when $r < \Gamma_{j,t}(O, I)$.*

Every global recovery induces a compatible family of block functionals. When that family is nonzero, fixing it lets the blocks minimize independently. When it is zero, leaving the target block requires an additional nonzero inner-dual equation. This explains the two terms and the separate $d(I^\perp)$ input: the ordinary zero-label minimum is zero. Section 4 proves that each finite minimum is attained by a coefficient-level lift.

Operational confinement: the second main result. For operational recovery, this extra external equation is dispensable. Let $\delta_j(O^\perp)$ be the least weight of an outer dual word nonzero at j, with value ∞ if there is none. Theorem 4.2 proves

$$\delta_j(O^\perp) > r + 1 \quad \Longrightarrow \quad \mathcal{M}_{j,T}^{(\le r)}(O \circ I) = \text{the embedded } \mathcal{M}_T^{(\le r)}(I),$$

where $\mathcal{M}$ denotes inclusion-minimal recovery supports of size at most r. The target-block part of any bounded recovery then recovers T on its own. Deleting the external helpers proves the assertion, without requiring the additive inner-dual bound for equation confinement. Equality of these minimal families transfers availability events under any law with the same local marginal, and transfers support-based capacity allocations. This target-touching condition is sufficient for operational transfer; unlike the equation threshold Γ, it does not characterize the first useful external repair.

The local cost has a classical interpretation. Puncturing the inner dual onto J gives D_P, and shortening it onto J gives K_P; shortening first requires the coordinates on P to vanish. The associated exact sequence identifies $D_P/K_P \cong W_P$. Its tth relative generalized Hamming weight $M_t(D_P, K_P)$ is the minimum helper union for recovering a t-dimensional subspace (Theorem 3.3). These weights, labelled costs, and physical supports retain different information, as Figure 1 records.

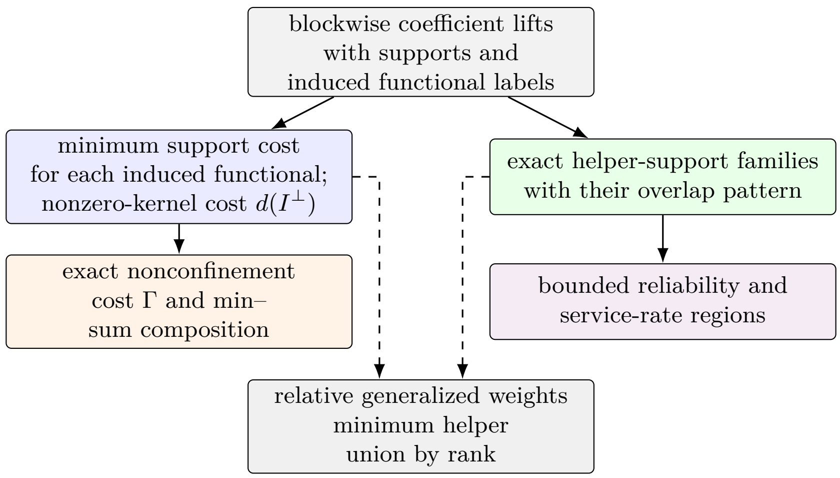

Figure 1: Blockwise lifts include nonzero functional fibres and zero-label inner equations. Outer compatibility and target normalization select recovery systems. Fibre minima and the separate nonzero-kernel cost determine numerical composition; retained minimizing lifts supply witnesses. Exact support families retain overlaps for reliability and allocation. Relative weights are the common minimum-cost projection (dashed arrows). Labelled support antichains refine both branches, composing before labels are forgotten and retaining nonnegative-price responses.

Space	Role in a recovery request
$\mathbb{F}_q^P, \mathbb{F}_q^J$	Coefficients on physical target and helper coordinates.
$U_P = \operatorname{im} G_P$	Message functionals supplied by target coordinates.
$T \le W_P \le U_P$	Requested functionals; W_P is the helper-recoverable part.
$D_P/K_P \cong W_P$	Helper coefficient lifts modulo target-free dual relations.
$L^* = \operatorname{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$	Functional labels passed from an inner block to its outer code.
$\mathrm{FD}(O) \le (L^*)^N$	Compatible tuples of block functionals annihilating the outer code.

Outer-distance collapse. Sufficient outer dual distance prevents a nonzero-label sector from improving the relative-weight cost plus one inner-dual perturbation. The exact characterization gives

$$d(O^\perp) \ge M_t(D_P, K_P) + d(I^\perp) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^\perp).$$

For a fixed radius r, the weaker condition $d(O^\perp) > r + 1$ reduces the exact criterion to

$$r < M_t(D_P, K_P) + d(I^\perp).$$

When this outer-distance condition and the displayed inequality both hold, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. If the outer dual distance tends to infinity, its required bound eventually holds at each fixed r; the displayed inner inequality is still required for this bijection.

The quotient-lifting law in Theorem 3.5 places recovery in a general nested-pair framework. Its support-antichain refinement retains resource choices, and all nonnegative helper prices determine the same minimal family as availability queries (Proposition 5.2). Scalar labelled costs give the specialized closure law in Theorem 4.4: the labelled prescribed-coset functions entering $\Gamma_{j,T}$ compose associatively by exact min–sum substitution. Storing an argmin in each attained fibre also propagates one coefficient-level witness; the numerical state itself contains only the fibre minima.

Appendix A gives finite tests for equivalence of numerical escape responses, with the original leaf boundary, target coordinates, and normalization fixed. Support overlap and reweighting require richer data.

The rank-one consequence is especially simple: a nonconfined recovery on a higher-dimensional target restricts to a nonconfined recovery on a target line without increasing its helper support. Hence confinement through a fixed radius holds at every recoverable rank if and only if it holds at rank one (Corollary 4.5).

Even identical image codes, relative-weight hierarchies, and unlabelled helper-support families can give different recovery costs with one fixed outer code. Proposition 4.3 realizes this over $\mathbb{F}_2 \subseteq \mathbb{F}_4$ by changing only the encoder's alignment with the functional labels. This is the compatibility issue in the running example, now with the unlabelled invariants held fixed.

Relation to existing viewpoints. Relative generalized Hamming weights and relative dimension/length profiles answer support and access questions for a nested pair [27, 24, 17]. Recovery sets and service-rate regions describe feasible helper choices and their use under capacities [18, 3]. These viewpoints explain the local cost and workload questions in the example. To pass from a local choice to an outer recovery equation, the interface must also retain the functional supplied by that choice.

The mechanism uses classical duality and min–sum elimination [2]. The recovery-specific content is the target normalization, the exact distinction between equation and minimal-support confinement, and the labelled lift law that reconstructs the selected coefficients. The separation results identify information lost by coarser summaries. Section 5.7 compares these statements with concatenated decoding, covering costs, and finite-interface minimization.

Section 5 describes executable recovery choices in Ergodis. Section 10 records verification scope and artifact references; measurements are in the software documentation.

Organization. Sections 2–4 develop the recovery model, local costs, equation and minimal-support transfer, and labelled composition. Section 5.1 carries one request through three encoding levels, a helper failure, and coefficient reconstruction; the rest of Section 5 gives the corresponding algorithms. Related work, separation examples, distance specializations, and geometric applications follow. Appendix A treats the secondary question of minimal numerical state.

2 Recovery sets and normalized equations

Let $C \leq \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r. A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \qquad (c \in C).$$

Its exact helper support is $\operatorname{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\leq r)}(C) = \{\operatorname{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \operatorname{wt}(w) \leq r+1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\leq r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r. Its inclusion-minimal members form the minimal recovery-set clutter. Here a clutter is a family of sets none of which contains another. Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^\perp : w_x = 1, \operatorname{wt}(w) \leq r+1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of $\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [39, 43, 21, 31].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \quad \iff \quad A \text{ contains a member of } \mathcal{H}_x^{(\leq r)}(C).$$

Under independent helper survival with probabilities (s_y), the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 6 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery system is a linear family whose image is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. We view its blocks as maps $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$. If $T \leq \operatorname{im} G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \qquad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P\alpha = \operatorname{id}_T, \qquad G_J\beta = -\operatorname{id}_T.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\operatorname{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\operatorname{supp}(\beta(T))|$. We call a target subspace helper- recoverable when it lies in

$$W_P = \operatorname{im} G_P \cap \operatorname{im} G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

3 Puncturing, shortening, and relative recovery costs

The local problem forgets the outer code and asks how many helpers realize a prescribed target combination. In the running example, the helper coefficient vectors $(1,0,0,0)$, $(0,1,0,0)$, and $(0,0,1,1)$ all realize u. Their differences lie in the kernel of the helper generator map. Passing to its quotient identifies the requested combination while leaving the cost of its different physical realizations to be minimized.

Continue with $G = (G_P \mid G_J)$, viewed as $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$, and set

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

$$K_P = \ker G_J, \qquad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair)**.** *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \tag{1}$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P. Its image is

$$G_J(D_P) = \operatorname{im} G_J \cap U_P = W_P,$$

so the restricted map is onto W_P. $\square$

The first isomorphism theorem applied to the restricted map gives $D_P/K_P \cong W_P$.

Proposition 3.2 (Canonical puncturing–shortening description)**.** *With puncturing and shortening taken onto the helper set J,*

$$D_P = \operatorname{punct}_J(I^\perp), \qquad K_P = \operatorname{short}_J(I^\perp). \tag{2}$$

Thus punct_J *restricts to J, while* short_J *first requires the coordinates on P to vanish and then restricts to J. These operations are dual to one another. Taking duals inside $\mathbb{F}_q^J$ gives*

$$D_P^\perp = \operatorname{short}_J(I), \qquad K_P^\perp = \operatorname{punct}_J(I). \tag{3}$$

Consequently the nested pair depends only on the code I and the split $E = P \sqcup J$; changing the row basis of a generator matrix leaves it unchanged.

Proof. A vector $a \in \mathbb{F}_q^J$ lies in $\operatorname{punct}_J(I^\perp)$ exactly when some $p \in \mathbb{F}_q^P$ satisfies $G_P p + G_J a = 0$, equivalently $G_J a \in \operatorname{im} G_P$, which is $a \in D_P$. It lies in $\operatorname{short}_J(I^\perp)$ exactly when $(0,a) \in I^\perp$, equivalently $G_J a = 0$, which is $a \in K_P$. For the first dual identity, a helper vector h is orthogonal to every punctured dual word exactly when $(0,h)$ is orthogonal to $I^\perp$, hence belongs to I. Thus $(\operatorname{punct}_J I^\perp)^\perp = \operatorname{short}_J I$. Apply this identity to $I^\perp$ and take orthogonal complements to obtain the second dual identity. This is the usual puncturing–shortening duality. $\square$

Thus a monomial relabeling of helpers gives the corresponding monomially equivalent pair, while a row-basis change does nothing to the pair itself.

For a subspace $A \leq \mathbb{F}_q^J$, write $\operatorname{supp} A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the tth relative generalized Hamming weight is

$$M_t(D,K) = \min\{|\operatorname{supp} L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t. Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [27]; standard treatments also cover ordinary Wei duality and computational forms [17, 38]. For nested code pairs in linear secret sharing, RDLPs and RGHWs already give exact information and reconstruction thresholds [24].

The following result is the standard nested-code support invariant applied to the canonical puncturing–shortening pair, with the normalization and helper support correspondence stated in recovery language.

Theorem 3.3 (Relative weights are exact recovery costs)**.** *Let $\ell = \dim W_P$ and $1 \le t \le \ell$. The minimum helper-union size among recovery systems for all helper-recoverable t-dimensional subspaces of target linear combinations, denoted $\mu_t(P)$, is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, after forgetting the target normalization, normalized recovery systems of recovered dimension t are in support-preserving correspondence with subspaces

$$L \le D_P, \qquad \dim L = t, \qquad L \cap K_P = 0,$$

while normalization on the chosen recovered subspace $T \le W_P$ is a linear section of G_P over T.

Proof. Let a normalized system recover a t-dimensional subspace $T \le W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\operatorname{id}_T$, the map β is injective, $L \le D_P$, and $L \cap K_P = 0$. Moreover $G_J L = T$, and the union of helpers in the system is exactly $\operatorname{supp} L$.

Conversely, let $L \le D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t-dimensional subspace $T = G_J L \le W_P$. Choose a linear section $\alpha : T \to \mathbb{F}_q^P$ with $G_P\alpha = \operatorname{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\operatorname{supp} L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \le D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L'. Then $\dim L = t$, $L \cap K_P = 0$, and $\operatorname{supp} L \subseteq \operatorname{supp} L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. ☐

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \bigl(\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)\bigr),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H.

Proposition 3.4 (Recovery interpretation of the relative profile)**.** *For $0 \le s \le |J|$,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \dim\bigl(W_P \cap \operatorname{span}(G_H)\bigr).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \ge t\}.$$

Thus K_s is the maximum number of independent target linear combinations recoverable from a fixed budget of s helpers.

Proof. Fix H. Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \operatorname{span}(G_H)$. Rank–nullity gives the first equality after maximizing over H. The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.3. □

Standard shortening–puncturing duality gives the complementary failure interpretation: the relative weights of $D_P^\perp \subsetneq K_P^\perp$ are the minimum numbers of helper failures that leave prescribed dimensions of W_P ambiguous [27, 17].

The strict inequalities

$$1 \leq M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that the optimal helper-union cost, minimized over all recoverable target subspaces of each dimension, strictly increases with that dimension. The cost need not strictly increase when a prescribed target subspace is enlarged: helpers $(u+v, v)$ recover u and (u, v) with the same two-helper union, although the best one-dimensional target costs one helper. These minima solve the single-block problem but do not yet provide recursive state: the next section restores the functional labels needed for exact concatenation. Only when an outer-distance condition removes every nonzero label does the answer collapse to $M_t(D_P, K_P) + d(I^\perp)$.

3.1 Prescribed quotient operations

The pair construction admits a useful general form. For finite linear codes $K \subsetneq D \leq \mathbb{F}_q^E$, let $\pi : D \to D/K$ be the quotient map. Given a finite $\mathbb{F}_q$-space T and a linear map $B : T \to D/K$, put

$$\kappa_T^{D/K}(B) = \min_{\substack{Y:T\to D \text{ linear} \\ \pi Y = B}} |\operatorname{supp} Y(T)|.$$

Thus a quotient label specifies an operation and a lift specifies its physical coordinates. Under $D_P/K_P \cong W_P$, the inclusion of a target subspace T has cost $\rho_T(I)$, the least helper-union size for recovering T in I: its helper lift has image under G_J equal to the requested target, and negating the helper coefficients gives the normalized dual equations. More generally, minimizing $\kappa_T^{D/K}(B)$ over injective B with $\dim T = t$ gives $M_t(D, K)$. Indeed an injective quotient image forces the lift image to be disjoint from K, and every t-dimensional subspace disjoint from K supplies such a map after choosing a basis.

Theorem 3.5 (Composition of quotient-labelled lifts)**.** *Let E_i be disjoint, let $K_i \subsetneq D_i \leq \mathbb{F}_q^{E_i}$, and write $\pi_i : D_i \to L_i = D_i/K_i$. For a nested pair $R_0 \subsetneq R_1 \leq \bigoplus_i L_i$, take inverse images inside $\bigoplus_i D_i$:*

$$\pi = \bigoplus_i \pi_i, \qquad D = \pi^{-1}(R_1), \qquad K = \pi^{-1}(R_0).$$

Then $D/K \cong R_1/R_0$. Writing $\bar{\pi} : R_1 \to R_1/R_0$, every linear $B : T \to R_1/R_0$ satisfies

$$\kappa_T^{D/K}(B) = \min_{\substack{Z:T\to R_1 \text{ linear} \\ \bar{\pi} Z = B}} \sum_i \kappa_T^{D_i/K_i}(\operatorname{pr}_i Z). \tag{4}$$

Minimizing lifts compose to a witness. Repeated substitution is associative.

Proof. The surjection $D \to R_1/R_0$ induced by π has kernel K, giving the asserted isomorphism. A lift $Y : T \to D$ determines $Z = \pi Y$ with $\bar{\pi} Z = B$. For fixed Z, its block maps Y_i are precisely the independent lifts of $\mathrm{pr}_i Z$. Their supports are disjoint, so their minimum costs add. All quotient maps admit linear sections; hence these minima are attained and their block witnesses assemble to a lift of B. This proves both inequalities in the formula. At further levels, either order of substitution minimizes over the same compatible intermediate maps and the same sum of leaf costs. $\square$

No common extension field is needed here. For heterogeneous injective encoders $e_i : V_i \hookrightarrow \mathbb{F}_q^{E_i}$ and a base-field-linear outer relation $O \leq \bigoplus_i V_i$, the interface is $e_i^* : \mathbb{F}_q^{E_i} \to V_i^*$ and compatibility is membership in $\mathrm{ann}(O)$. Orthogonality to all encoded outer words proves this directly. Target-normalized fibres retain their target constraints and exclude target coordinates from the cost. To identify a prescribed target with the full alphabet V_j, the projection $O \to V_j$ must be surjective; a nonzero projection alone is insufficient. With nonzero T and at least two blocks, the zero-functional nonconfinement contribution is $\rho_{j,T} + \min_{i \neq j} d(I_i^{\perp})$, by adjoining a minimum dual word in the cheapest external block. Extension-field scalar and projective actions used below are additional structure of that specialization.

Available quotient directions. For a coordinate set $A \subseteq E$, the quotient directions realizable entirely on A form

$$L(A) = \pi(D \cap \mathbb{F}_q^A), \qquad \dim L(A) = \dim(D \cap \mathbb{F}_q^A) - \dim(K \cap \mathbb{F}_q^A).$$

Here $\mathbb{F}_q^A$ is embedded by zero-extension. The formula is rank–nullity for the restricted quotient map. A prescribed B has a lift on A exactly when $B(T) \subseteq L(A)$, since the restricted surjection admits a linear section. This retains the directions exposed on A, as well as their maximum possible rank; minimizing ordinary words of D would lose the requirement that the quotient label be nonzero.

4 Exact confinement under concatenation

Two confinement results govern the section. Classifying recovery systems by their outer functional labels gives the exact equation escape cost (Theorem 4.1). Deleting dispensable external coefficients instead gives minimal-support transfer under a target-touching distance hypothesis (Theorem 4.2). The latter transfers availability and support-based allocations even when equations can escape. Finally, the full label-indexed costs pass to the next concatenation level by an associative min–sum law.

Represented inner code. Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \leq \mathbb{F}_q^E$ be an $\mathbb{F}_q$-linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. Whenever the outer alphabet L occurs below, I denotes the represented inner code (I, ι), not only its image subspace. In particular, $O \circ I$, Φ_I, $\lambda_{I,T}$, $\mu_{I,P,T}$, and $\Gamma_{j,T}(O, I)$ depend on ι. The earlier pair $K_P \subseteq D_P$ depends only on the image code and target/helper split. Proposition 4.3 shows that this distinction cannot be suppressed under concatenation.

For an L-linear outer code $O \leq L^N$, write

$$O \circ I = \{(\iota(a_1), \ldots, \iota(a_N)) : (a_1, \ldots, a_N) \in O\}.$$

Each block has a label map

$$\underbrace{\mathbb{F}_q^E}_{\text{inner coefficients}} \xrightarrow{\Phi_I} \underbrace{L^*}_{\text{functional label}}, \qquad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. The chosen message coordinates identify each earlier generator column with the functional $a \mapsto \iota(a)_e$; equivalently, it is the unique trace coefficient b_e with $\iota(a)_e = \mathrm{Tr}_{L/\mathbb{F}_q}(b_e a)$. We transport T and $G_P\alpha = \mathrm{id}_T$ through this same identification. Two coefficient vectors receive the same label precisely when they differ by an inner dual word, because $\ker \Phi_I = I^\perp$. The prescribed-coset costs below record the cheapest coefficient support in each fibre of this map.

Outer compatibility. A block vector $y = (y_1, \dots, y_N)$ belongs to $(O \circ I)^\perp$ precisely when

$$(\Phi_I(y_1), \dots, \Phi_I(y_N)) \in \mathrm{FD}(O), \tag{5}$$

where

$$\mathrm{FD}(O) = \{(\beta_j) \in (L^*)^N : \textstyle\sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Write $\beta_j(a) = \mathrm{Tr}_{L/\mathbb{F}_q}(b_j a)$. The functional-dual condition becomes

$$\mathrm{Tr}_{L/\mathbb{F}_q}\left(\sum_j b_j a_j\right) = 0 \qquad \text{for every } (a_j) \in O.$$

Because O is L-linear, the same holds after multiplying (a_j) by every $\lambda \in L$. Nondegeneracy of the trace pairing then forces $\sum_j b_j a_j = 0$. Hence $(b_j) \in O^\perp$, and the converse is immediate. This identification preserves block support, so the support distance of $\mathrm{FD}(O)$ is $d(O^\perp)$.

Confinement and target normalization. Fix a target block and a nonzero target-message subspace $T \le U_P$. A system is *confined* if every one of its equations is supported in that target block. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code, with value ∞ when no such system exists. Thus $\rho_T(I) < \infty$ exactly when $T \le W_P$. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P\alpha = \mathrm{id}_T$ under the fixed inner-message identification. This fixes a normalization condition, not a particular section α: the admissible sections vary in the minimization. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^\perp) > 1$ guarantees surjectivity: an L-linear coordinate image is either 0 or L, and a zero image would put a weight-one word in $O^\perp$.

For an $\mathbb{F}_q$-linear map $B : T \to L^*$, define

$$\lambda_{I,T}(B) = \min_{\substack{Y:T\to\mathbb{F}_q^E \text{ linear} \\ \Phi_I Y = B}} |\operatorname{supp} Y(T)|. \tag{6}$$

For the target block, define

$$\mu_{I,P,T}(B) = \min_{\substack{\alpha:T\to\mathbb{F}_q^P,\ \beta:T\to\mathbb{F}_q^J \text{ linear} \\ G_P\alpha = \mathrm{id}_T \\ \Phi_I(\alpha,\beta)=B}} |\operatorname{supp} \beta(T)|. \tag{7}$$

An empty minimum is ∞. After choosing a basis of T, (6) is the minimum union support of representatives of prescribed cosets of $I^\perp$. This is the fixed-instance quantity underlying generalized coset weights and generalized covering radii [44, 12]. The target version retains the normalization required by the recovery problem. In particular,

$$\mu_{I,P,T}(0) = \rho_T(I). \tag{8}$$

For each equation parameter $t \in T$, the value $B(t)$ is exactly the outer functional label induced by that equation in the block.

Let

$$\mathcal{B}_T(O) = \mathrm{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

For $B \in \mathcal{B}_T(O)$, write $B = (B_h)_{h=1}^N$ with $B_h : T \to L^*$. Assume $N \geq 2$, fix a target block j whose coordinate projection $O \to L$ is nonzero, hence surjective, and put

$$\begin{aligned}\Gamma_{j,T}(O,I) = \min\Bigg\{&\mu_{I,P,T}(0) + d(I^\perp), \\ &\min_{0\neq B\in\mathcal{B}_T(O)} \Bigg(\mu_{I,P,T}(B_j) + \sum_{h\neq j} \lambda_{I,T}(B_h)\Bigg)\Bigg\}.\end{aligned} \tag{9}$$

Theorem 4.1 (Exact nonconfinement cost from labelled coset supports)**.** *Under the hypotheses above, $\Gamma_{j,T}(O,I)$ is the least helper-union size of a nonconfined normalized recovery system for T in $O \circ I$, with value ∞ if no such system exists. Thus every cost-$\leq r$ system for T is confined if and only if*

$$r < \Gamma_{j,T}(O,I). \tag{10}$$

For $1 \leq t \leq \ell = \dim W_P$, set

$$\Gamma_{j,t}(O,I) = \min_{\substack{T\leq W_P\\ \dim T = t}} \Gamma_{j,T}(O,I). \tag{11}$$

Then every cost-$\leq r$ system for every helper-recoverable t-dimensional target subspace is confined if and only if $r < \Gamma_{j,t}(O,I)$. Below either threshold, restriction and zero-extension are inverse bijections on normalized systems and preserve exact helper supports.

Proof. Write a recovery system as an $\mathbb{F}_q$-linear map $Y : T \to (O \circ I)^\perp$ and let Y_h be its block maps. By (5),

$$B_h = \Phi_I Y_h$$

defines a linear map $B : T \to \mathrm{FD}(O)$. Conversely, compatible linear lifts of the block maps of any such B give a map into the concatenated dual. At the target block, compatibility with the chosen target space is exactly the constraint in (7).

Suppose first that $B \neq 0$. No nonzero element of $\mathrm{FD}(O)$ is supported only at j: testing such an element against outer words whose jth coordinate ranges over L would force its jth functional to vanish. Hence some block map B_h with $h \neq j$ is nonzero, and every lift is nonconfined. Distinct inner blocks have disjoint helper coordinates, so the minimum helper union for this fixed B is

$$\mu_{I,P,T}(B_j) + \sum_{h\neq j} \lambda_{I,T}(B_h).$$

The block lifts can be chosen independently, so this lower bound is attained.

If $B = 0$, every block map takes values in $I^\perp = \ker \Phi_I$. The target block costs $\rho_T(I) = \mu_{I,P,T}(0)$. Nonconfinement requires a nonzero map $T \to I^\perp$ in another block. Its image has union support at least $d(I^{\perp})$, and equality is attained by $u \mapsto f(u)v$, where $0 \neq f \in T^*$ and v is a minimum-weight word of $I^{\perp}$.

The zero and nonzero functional sectors are exhaustive, proving (9) and (10). Minimizing over t-dimensional T proves the ranked statement. Below the first nonconfined cost, every global system has zero coefficient maps in all blocks outside j. Restriction therefore preserves its target normalization and helper support. Conversely, zero-extending an inner dual system remains orthogonal to every concatenated word, so the two operations are inverse with coefficients and supports unchanged. □

The running example: equations versus useful repairs. For the code in the introduction, $G = (e_1 \mid e_1, e_1, e_2, e_1 + e_2)$. Its inner-dual distance is two: two repeated columns give a weight-two relation, and no column is zero. The local recovery cost for u is one. Identify its message space with $L = \mathbb{F}_4$ and take $O = L^N$, $N \geq 2$. There are no nonzero outer-dual labels, so the exact formula gives $\Gamma_{j,1} = 1 + 2 = 3$. A cost-three nonconfined equation adds the relation $u + u = 0$ in another block to the local recovery of u. Deleting those two external coefficients leaves the original repair.

Nevertheless every minimal repair remains in its own block at every radius: the outer messages are independent, so external coefficients cannot supply any part of the target functional. The three local choices remain $\{1\}$, $\{2\}$, and $\{3, 4\}$. This example exhibits the zero sector of the exact formula and the difference between its coefficient threshold and the operational theorem below. The nonzero-sector alignment obstruction is exhibited separately in Proposition 4.3.

4.1 Minimal supports and operational transfer

An external equation can violate coefficient confinement without supplying a new recovery option. To separate these questions, let $\mathcal{M}^{(\leq r)}_{j,T}(O \circ I)$ be the inclusion-minimal helper supports recovering T with at most r helpers. A set recovers T when some normalized system uses only that set. Define the target-touching distance

$$\delta_j(O^{\perp}) = \min\{\mathrm{wt}(u) : u \in O^{\perp},\ u_j \neq 0\},$$

with empty minimum ∞.

Theorem 4.2 (Minimal-support confinement)**.** *Let $O \leq L^N$ be L-linear, let its projection onto block j be surjective, and let $0 \neq T \leq U_P$. If $\delta_j(O^{\perp}) > r + 1$, then*

$$\mathcal{M}^{(\leq r)}_{j,T}(O \circ I) = \{\{j\} \times H : H \in \mathcal{M}^{(\leq r)}_{T}(I)\}.$$

No inequality involving $\rho_T(I) + d(I^{\perp})$ is required. Consequently bounded recovery success has the same probability for every global helper-availability law with the prescribed target-block marginal.

Proof. Let Y be a normalized system with helper support H, $|H| \leq r$, and put $B_h = \Phi_I Y_h$. If $B_j \neq 0$, choose $v \in T$ with $B_j(v) \neq 0$. The trace identification sends $B(v)$ to an outer dual word touching j. Its other nonzero blocks each contain a helper in H, so its weight is at most $r + 1$, a contradiction. Thus $B_j = 0$, and the target-block portion of Y is an inner-dual system with the same target normalization. Deleting all external coefficients preserves recovery and only decreases support. Every global minimal support is therefore local. Conversely, a proper globally recovering subset of a local minimal support would, by this same restriction argument, contradict local minimality. Zero-extension supplies the other inclusion. For each available set, success means containing one of these minimal supports; the two success indicators therefore agree pointwise after restriction to the target block. Taking expectations proves the probability assertion without an independence assumption. □

The first new external repair has its own threshold. Let $\varepsilon_{j,T}$ be the least size of a globally recovering helper set H whose portion in block j does not recover T locally. Equivalently, it is the least size of a global minimal support using an external helper: minimize inside H for one direction; a local recovery inside a minimal support contradicts minimality for the other. The theorem gives $\varepsilon_{j,T} \geq \delta_j(O^\perp) - 1$ when the latter is finite, and $\varepsilon_{j,T} = \infty$ when $\delta_j(O^\perp) = \infty$. Its value cannot be obtained simply by deleting the zero sector from $\Gamma_{j,T}$: a nonzero-labelled lift can also contain a local recovery. For $O = L^N$, every minimal recovery support is local at every radius, although adjoining an external inner-dual word gives the finite equation threshold $\rho_T(I) + d(I^\perp)$ whenever T is locally recoverable.

4.2 Repeated concatenation

The ordinary prescribed-coset costs retain the intermediate labels needed at the next concatenation level. Target-normalized composition also retains the part of each intermediate label already supplied by target coordinates. In schematic form,

$$\text{cost of output label} = \min_{\text{compatible intermediate labels}} \sum_{\text{inner blocks}} \text{inner realization cost.}$$

The eliminated variables are the intermediate labels shared by adjacent levels. The resulting formulas are instances of variable elimination over the min-sum semiring [2]. This is also the algebraic form of passing symbol-indexed inner costs to an outer decoder, as in soft and generalized-concatenated decoding [20, 6].

The zero-functional nonconfinement cost alone is not compositional. The next separation holds the underlying binary code and all its unlabelled recovery data fixed.

Proposition 4.3 (Unlabelled data do not determine finite composition)**.** *Over $\mathbb{F}_2 \subseteq \mathbb{F}_4$, there are two represented inner encoders with the same image code, dual code, target/helper nested pair, complete relative-weight hierarchy, dual distance, unlabelled normalized helper-support family, and zero-functional nonconfinement costs for every recoverable target subspace, but one fixed outer represented code gives different exact nonconfinement costs.*

Proof. Write the nonzero elements of $\mathbb{F}_4$ as $1, \omega, \omega^2 = 1 + \omega$ and take generator columns

$$I_1 : (1, 1, \omega), \qquad I_2 : (1, 1, \omega^2),$$

with the first coordinate targeted. Relative to the binary basis $(1, \omega)$ of $\mathbb{F}_4$, each displayed field element is a two-dimensional binary generator column. Both generator matrices have binary row space

$$\{(x, x, y) : x, y \in \mathbb{F}_2\}$$

and dual $\langle(1, 1, 0)\rangle$. Thus both have dual distance 2, $K_P = 0 \subseteq D_P = \langle(1, 0)\rangle$, $M_1 = 1$, and the same unique minimal helper support. Their recoverable target space is one-dimensional, so their complete zero-functional profile is the common value $1 + 2 = 3$.

For this table identify a column c with the functional $a \mapsto \mathrm{Tr}(\omega^2 ca)$. This is the coordinate-dot-product pairing in the basis $(1, \omega)$; the corresponding trace coefficient is $\omega^2 c$. The common rescaling preserves every outer L-linear dual subspace. In this column-label order $(1, \omega, \omega^2)$, the ordinary coset costs are $(1, 1, 2)$ for I_1 and $(1, 2, 1)$ for I_2; the normalized target costs are respectively $(0, 2, 1)$ and $(0, 1, 2)$. Take the fixed outer code whose functional dual is the line spanned by $(1, \omega)$. Its nonzero multiples have target label s and external label $s\omega$. The complete calculation is

s	s	$s\omega$	$\mu_1(s)$	$\lambda_1(s\omega)$	total	$\mu_2(s)$	$\lambda_2(s\omega)$	total
1	1	ω	0	1	1	0	2	2
ω	ω	ω^2	2	2	4	1	1	2
ω^2	ω^2	1	1	1	2	2	1	3

Equation (9) therefore gives exact costs 1 and 2. Only the alignment of costs with the intermediate functional labels has changed. $\square$

Let $\mathbb{F}_q \subseteq L \subseteq M$ be a tower of finite fields. Let B be an $\mathbb{F}_q$-linear represented code with message space L, and let A be an L-linear represented code with message space M. The composite $A \circ B$ replaces each coordinate of an A-word by its represented B-block. After identifying a functional with its unique trace coefficient, write their dual restriction maps as

$$\phi_B : \mathbb{F}_q^{E_B} \longrightarrow L, \qquad \phi_A : L^{E_A} \longrightarrow M.$$

For an $\mathbb{F}_q$-space T and $b : T \to L$, put

$$\Lambda_{B,T}(b) = \min_{\phi_B Y = b} |\operatorname{supp} Y(T)|.$$

Define $\Lambda_{A,T}$ and $\Lambda_{A\circ B,T}$ similarly. If $J \subseteq E_B$, let $\Lambda_{B,J,T}$ denote the same minimum with Y supported on J and ϕ_B restricted to those coordinates. Thus ϕ is the trace-coordinate form of Φ, and Λ is the corresponding ordinary prescribed-coset cost λ. The change of letters only distinguishes the two levels in the following substitution.

For the recovery-specific form, suppose that a target set in $A \circ B$ meets block e in $P_e \subseteq E_B$. Put $J_e = E_B \setminus P_e$ and $U_{P_e} = \operatorname{im} \phi_{B,P_e}$. If $T \leq M$, let $\iota_T : T \hookrightarrow M$ be the inclusion. The variables below have the following roles:

U_e	intermediate contribution supplied by target coordinates in block e
X_e	total intermediate functional label required in block e
$X_e - U_e$	label that the helper coordinates in block e must realize.

Theorem 4.4 (Min–sum closure of prescribed-coset support costs)**.** *For every $\mathbb{F}_q$-linear map $c : T \to M$, the ordinary prescribed-coset cost is*

$$\Lambda_{A\circ B,T}(c) = \min_{X:T\to L^{E_A}} \min_{\phi_A X = c} \sum_{e\in E_A} \Lambda_{B,T}(X_e). \tag{12}$$

For $T \leq M$ and the target split above, the target-normalized recovery cost is

$$\mu_{A\circ B,P,T}(c) = \min_{\substack{U,X:T\to L^{E_A} \\ U_e(T)\subseteq U_{P_e}\ (e\in E_A) \\ \phi_A U = \iota_T,\ \phi_A X = c}} \sum_{e\in E_A} \Lambda_{B,J_e,T}(X_e - U_e). \tag{13}$$

Both formulas iterate associatively through a finite tower: either parenthesization minimizes over the same intermediate labels and target contributions and sums the same leaf costs. Storing minimizing target and helper lifts propagates one coefficient-level witness; retaining every witness requires the full lift sets rather than only the numerical minima.

Proof. A leaf-level lift $Y : T \to \mathbb{F}_q^{E_A \times E_B}$ determines intermediate maps $X_e = \phi_B Y_e$. Its support is the disjoint union of its supports in the inner blocks. At fixed X, the block lifts are independent and attain their minima separately, which proves (12).

For the target-normalized formula, lift U_e to an actual target map $a_e : T \to \mathbb{F}_q^{P_e}$. The condition $\phi_A U = \iota_T$ is exactly target normalization. Once X_e is fixed, the helper contribution must lift $X_e - U_e$ through ϕ_{B,J_e}, independently in each block. This proves (13); storing the displayed lifts gives the stated witness semantics. At three or more levels, either parenthesization eliminates the same intermediate arrays and target contributions, proving associativity. □

Sharp scalar envelopes. The full labelled function gives simple bounds when only its smallest and largest nonzero values are retained. When $T \neq 0$, set

$$\delta_{B,T} = \min_{0 \neq b : T \to L} \Lambda_{B,T}(b), \qquad R_{B,T} = \max_{0 \neq b : T \to L} \Lambda_{B,T}(b).$$

For $c \neq 0$, the exact formula has the sharp coarsening

$$\delta_{B,T} \Lambda_{A,T}(c) \leq \Lambda_{A \circ B,T}(c) \leq R_{B,T} \Lambda_{A,T}(c). \tag{14}$$

Neither constant can be improved without further hypotheses. Indeed, every nonzero intermediate coordinate map costs between $\delta_{B,T}$ and $R_{B,T}$. Minimizing the number of nonzero intermediate coordinates proves the bounds; a one-coordinate outer realization attaining either extremum proves sharpness.

Dual-distance recursion. The same zero/nonzero-label split gives

$$d((A \circ B)^\perp) = \min \left\{ d(B^\perp), \min_{0 \neq z \in A^\perp} \sum_{e \in E_A} \Lambda_{B,\mathbb{F}_q}(u \mapsto u z_e) \right\}. \tag{15}$$

A word in $(A \circ B)^\perp$ either has zero intermediate label, when a minimum word of $B^\perp$ in one block is optimal, or has a nonzero label $z \in A^\perp$, when the block representatives minimize independently. This proves (15).

Tower consequence. Substituting these recursively computed costs into (9) gives the exact nonconfinement cost through any finite tower, with the zero and nonzero functional sectors kept separate at every level.

The trace identifications in these formulas are compatible with the tower. Write $\iota_A : M \to A \leq L^{E_A}$ for the represented encoder of A. For $v \in T$ and $m \in M$, trace transitivity gives

$$\sum_e \mathrm{Tr}_{L/\mathbb{F}_q}\big(X_e(v)\, \iota_A(m)_e\big) = \mathrm{Tr}_{M/\mathbb{F}_q}\big(\phi_A(X(v))m\big),$$

and nondegeneracy of the trace pairing identifies the composite restriction map with $\phi_A \circ \phi_B^{E_A}$, where $\phi_B^{E_A}$ applies ϕ_B blockwise.

Simultaneous confinement in every dimension.

Corollary 4.5 (Rank-one criterion for complete bounded transfer)**.** *For every target block j with nonzero outer projection,*

$$\min_{0 \neq T \leq W_P} \Gamma_{j,T}(O, I) = \Gamma_{j,1}(O, I). \tag{16}$$

For every integer $r \geq 0$, the following are equivalent:

1. *every cost-$\leq r$ system on every helper-recoverable target line is confined;*
2. *every cost-$\leq r$ system on every nonzero helper-recoverable target subspace is confined; and*
3. $r < \Gamma_{j,1}(O, I)$.

Under these conditions, restriction and zero-extension are inverse bijections simultaneously for every nonzero $T \leq W_P$, preserving every coefficient and exact helper support through radius r.

Consequently, any statistic defined solely from these bounded systems and invariant under this bijection is preserved. In particular, the bounded reliability laws and the capacity and scheduling regions defined later from their coefficients or supports agree. Targetwise and rankwise minimum costs also agree after truncation $c \mapsto \min\{c, r+1\}$ (with $\min\{\infty, r+1\} = r+1$). If every relevant inner minimum is at most r, then the untruncated minima agree as well.

Proof. If a system on $0 \neq T \leq W_P$ is nonconfined, some external block map is nonzero on a vector $u \in T$. Restriction to $\langle u \rangle$ remains nonconfined and cannot enlarge its helper union. This proves the displayed equality, and Theorem 4.1 gives the three equivalent conditions. When they hold, every bounded system is confined, so restriction deletes only zero external blocks and zero-extension is its inverse. The coefficient, support, reliability, capacity, and scheduling assertions follow from this bijection. For a minimum helper-union cost, either the inner minimum is at most r and the bijection gives equality, or both inner and concatenated minima exceed r; this is precisely equality after truncation at $r + 1$. □

5 Exact recovery optimization

The labelled composition law makes an exact local calculation reusable inside larger recovery problems. Ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) embodies this construction: it retains the intermediate functionals needed to compose choices and the lifts needed to execute them. Its recovery interfaces connect three operations. A compiled table answers prescribed-functional queries; retained support alternatives allow failures, helper prices, and capacity constraints to select different repairs; stored coefficients reconstruct the selected repair in the original code. The system is developed at `tavisrudd/ergodis`.

We develop the numerical recursion, its support-valued refinement, and a fixed-query width bound, then apply the interface to bilinear worker tasks. The cost counts participating scalar helper coordinates.

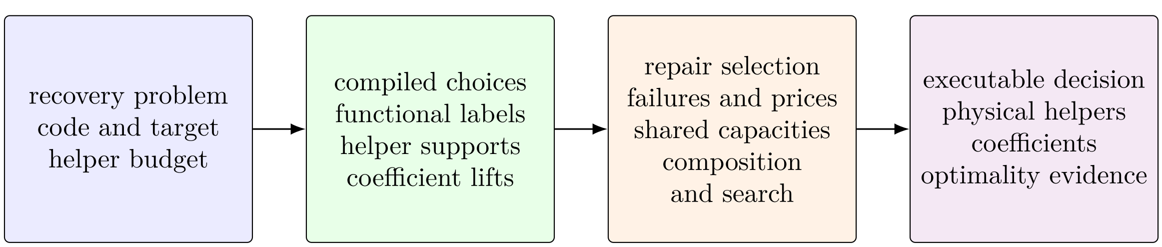

Figure 2: Ergodis separates the source model, admitted representation, optimization, and evidence contract. Functional labels preserve composition; support alternatives preserve overlap and helper-price queries. Stored lifts reconstruct feasible decisions, while optimality needs a matching checked bound or complete search justification. Reuse under updates or new readouts requires admission for the chosen observation. The mathematical results do not depend on executing this pipeline.

5.1 A complete three-level repair

Two repetitions and one parity relation give a small hierarchy in which a failed helper changes the outer choice. Over $\mathbb{F}_2$, let

$$B : z \longmapsto (z, z), \qquad A : z \longmapsto (z, z), \qquad C : (u, v) \longmapsto (u, v, u + v).$$

The code $C \circ (A \circ B)$ has coordinates y_{gij}, where $g \in \{a, b, c\}$ names the outer symbol, $i \in \{1, 2\}$ its A copy, and $j \in \{1, 2\}$ its B copy. Thus all four coordinates with outer index a, b, c equal $u, v, u+v$, respectively. Target $y_{a11} = u$; the request space is $T = \mathbb{F}_2$, and its target coefficient is fixed to 1. The same labels may be regarded as linear maps from T, determined by their values at 1.

Give the eleven helpers the following nonnegative prices; the dash marks the target, which is not a helper:

	y_{g11}	y_{g12}	y_{g21}	y_{g22}
a	–	1	4	4
b	1	4	4	4
c	1	4	4	4

At an ordinary B leaf with prices (p_1, p_2), the restriction map is $(h_1, h_2) \mapsto h_1 + h_2$. Its labelled price table is $\lambda(0) = 0$, $\lambda(1) = \min\{p_1, p_2\}$. At the target leaf, the total functional label x satisfies $1 + h_2 = x$, so $\mu(0) = p_{a12}$ and $\mu(1) = 0$. This is the target-normalization constraint in (13): the fixed target contribution is 1, and helpers supply $x - 1$.

At the A level, child labels add to the parent label x. For example,

$$\mu_a(x) = \min_{x_1+x_2=x} \big(\mu_{a1}(x_1) + \lambda_{a2}(x_2)\big).$$

The resulting tables, before and after helper y_{a12} fails, are

	initial		y_{a12} unavailable	
x	0	1	0	1
$\mu_a(x)$	1	0	4	0
$\lambda_b(x)$	0	1	0	1
$\lambda_c(x)$	0	1	0	1

Only the target leaf table changes: its zero-label entry becomes ∞. The value $\mu_a(0) = 4$ is then attained by child labels $(1, 1)$, lifting the second label to y_{a21} or y_{a22}.

At the top, orthogonality to $(u, v, u + v)$ requires $X_a = X_b = X_c = \varepsilon$. Hence the full repair price is

$$\min_{\varepsilon\in\mathbb{F}_2} \big(\mu_a(\varepsilon) + \lambda_b(\varepsilon) + \lambda_c(\varepsilon)\big) = \begin{cases} \min\{1, 2\} = 1, & \text{initially,} \\ \min\{4, 2\} = 2, & \text{after the failure.} \end{cases}$$

Initially the minimizing outer label is $(0, 0, 0)$ and the lifted equation is $y_{a11} + y_{a12} = 0$. After failure it is $(1, 1, 1)$; each A block uses child labels $(1, 0)$. In block a the target alone realizes the first label; in blocks b, c choose helper coefficients $(1, 0)$ at the first B leaf. Backtracking therefore returns

$$y_{a11} + y_{b11} + y_{c11} = 0, \qquad u + v + (u + v) = 0.$$

The two branches exhaust the top compatibility condition, and each local table is attained, proving both optima. Reusing only the initial minimizing witness would miss the second repair. The linear interfaces stay fixed; availability changes the local table and its ancestor messages. Here the initial and replacement helper unions have sizes one and two, respectively; the objective in the calculation is their stated total price.

5.2 The labelled dynamic program

We first state the algorithm in flattened $\mathbb{F}_q$-linear coordinates. Let T have dimension t, let the outer message space have dimension m, and let the inner label space have dimension s. For the ith outer coordinate, write

$$A_i : \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^s) \longrightarrow \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$$

for its contribution to the outer restriction map, and let $\lambda_i(b)$ be the prescribed-coset support cost of the inner label b. Different inner blocks may have different cost functions. For chosen block labels $b_1, \ldots, b_N$, the total outer label is $\sum_i A_i(b_i)$; the dynamic program enforces that this sum equals the requested label c.

Proposition 5.1 (Exact hierarchical recovery optimizer)**.** *For $c \in \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$, define*

$$D_0(c) = \begin{cases} 0, & c = 0, \\ +\infty, & c \neq 0, \end{cases} \qquad D_i(c) = \min_b \big(D_{i-1}(c - A_i b) + \lambda_i(b)\big). \tag{17}$$

Then $D_N(c)$ is the exact prescribed-coset support cost of the concatenated code at label c. Storing a minimizing label b and predecessor state at each finite entry returns an exact block-labelled recovery witness, not only its scalar cost. The same recursion applies to the target-normalized formula after adjoining its target-image labels to the state.

If every label is enumerated, put $Q = q^{mt}$ and $S = q^{st}$. The direct table algorithm uses $O(NQS)$ min–sum transitions, $O(Q)$ working cost memory, and $O(NQ)$ predecessor entries when one minimizing witness is retained for each finite table entry.

Proof. After i blocks, $D_i(c)$ is the least sum of inner support costs among labels $(b_1, \ldots, b_i)$ satisfying $\sum_{j \leq i} A_j b_j = c$. This invariant is true at $i = 0$. Conditioning on the last label b_i gives exactly (17), so induction proves the claim. The leaf supports lie in disjoint blocks, hence their costs add. Backtracking the stored minimizers reconstructs the labels; the inner prescribed-coset witnesses then reconstruct the coefficient-level recovery system. There are Q states and S candidate labels per block, which gives the stated bounds. The target-normalized case is the same invariant applied to the paired labels in (13). ☐

The predecessor bound counts entries; local coefficient lifts have their own storage cost. Retaining every tied transition can require $O(NQS)$ entries, and enumerating all complete minimizing recoveries incurs their output cost. Support alternatives require the families below, including larger minimal supports that can become useful after a failure or price change.

The implementation also removes unreachable labels and merges states only when an explicit expansion map identifies each reduced choice with original block labels of the same cost. Backtracking through that map recovers an original-instance witness. Quotient coordinates, projectively equivalent columns, and repeated block types are instances of this implementation rule; they are not additional claims of Proposition 5.1. Proposition A.3 supplies a separate theorem-backed reduction for composition: every target-rank-t witness factors through the L-span generated by its image, whose dimension is at most t. An exact search may therefore cache that generated span and discard states of larger dimension without losing an optimum or witness. This optional reduction uses the bounded outer-test theorem; the elementary dynamic program above does not require it.

The resulting fibre tables, together with the separately propagated nonzero-kernel cost in (15), supply the finite confinement formula. The optimizer can therefore emit a replayable recovery witness when a threshold fails, or a complete exhausted-state certificate when the requested label is unreachable.

5.3 Interfaces for resource choices and availability

In the running example, a helper budget of two and nonnegative prices give the exact repair cost

$$\min\{p_1, p_2, p_3 + p_4\}.$$

A unit-cost table alone returns one, but does not retain this function. The same three supports give the availability event 1 survives $\vee$ 2 survives $\vee$ (3 and 4 survive). Retain these alternatives within each functional label. For each ordinary label b, let

$$\mathcal{A}_{i,T}(b) = \min_{\subseteq}\{\operatorname{supp} Y(T) : \Phi_i Y = b\},$$

where $\min_{\subseteq}$ removes strict supersets and the empty fibre gives the empty family. The family $\{\varnothing\}$ is distinct from the empty family. In the notation of the flattened optimizer, the exact substitution is

$$\mathcal{A}_T^{\mathrm{out}}(c) = \min_{\subseteq}\left\{\bigcup_i H_i : \sum_i A_i b_i = c, \quad H_i \in \mathcal{A}_{i,T}(b_i)\right\}. \tag{18}$$

Indeed every lift decomposes into these block fibres; replacing one block support by a contained support with the same label preserves compatibility. Conversely every displayed choice lifts. The same argument applies to Theorem 3.5, and to target normalization with its existing constraints. Retain a coefficient witness for each surviving support. For additive nonnegative resource loads one may instead retain a Pareto frontier within each label: coordinatewise domination is preserved by addition. This is safe only for the resource observations represented by that load vector. The numerical contextual quotient in Corollary A.4 does not automatically preserve these richer observations.

Proposition 5.2 (Supports, prices, and availability)**.** *Fix a finite labelled helper set E, a radius, and its minimal recovery family $\mathcal{M}$. The following data determine one another: $\mathcal{M}$; the success indicator $F(A) = \mathbf{1}\{\exists H \in \mathcal{M} : H \subseteq A\}$ on every $A \subseteq E$; the price function*

$$\pi_{\mathcal{M}}(p) = \min_{H \in \mathcal{M}} \sum_{h \in H} p_h \quad (p \in \mathbb{R}^E_{\geq 0});$$

and the reliability for every independent helper survival vector in $[0,1]^E$. Empty price minima have value ∞.

Proof. The family gives both displayed functions and its reliability by summing $F(A)$ against the probability of each available set. The minimal successful sets of F recover $\mathcal{M}$. Prices zero on A and one outside A satisfy $\pi_{\mathcal{M}}(p) = 0$ exactly when $F(A) = 1$, including the empty-family case. Finally deterministic survival vectors recover every $F(A)$ from reliability. These implications prove the assertion. □

This equivalence concerns information, not the number of oracle calls. Extracting reliability can require exponentially many availability queries. The running example also shows why summing repair probabilities fails. If its four helpers survive independently with probability 1/2, failure requires losing helpers 1 and 2 and at least one of helpers 3 and 4. Hence reliability is $1-(1/2)^2(1-1/4) = 13/16$. Adding the three individual repair probabilities instead gives $1/2 + 1/2 + 1/4 = 5/4$. One must evaluate the existential event F, for example by inclusion–exclusion or a Boolean representation that retains overlap. Factoring probabilities is valid on independent variable sets; adding alternative probabilities is valid on disjoint events.

Fractional service optimization. Retain the radius as a separate constraint and evaluate $\pi_a(p)$ over supports of size at most r. For the fractional objective $\max \sum_a w_a \lambda_a$ under helper capacities c_h, the full dual is

$$\min_{p \geq 0} \sum_h c_h p_h, \qquad \sum_{h \in R} p_h \geq w_a \quad (a \in \mathcal{A},\ R \in \mathcal{M}_a^{(\leq r)}).$$

Thus solving a restricted service LP, pricing every demand, and adding a witness whenever $w_a - \pi_a(p) > 0$ yields column generation. If the restricted primal and dual have equal finite values and no price violation remains, their value is the full optimum: the restricted primal is feasible for the full problem and the prices give a matching full dual bound. This is a fractional certificate. Integral scheduling needs its own exact search or bounds. Support-family equality transfers an integral feasible region, but an LP optimum need not be an integral optimum.

5.4 Width of a fixed query

The ambient syndrome dimension is a pessimistic state bound. Fix c and write $A_i b = a_i \circ b$, where $a_i : L_i \to W$ is linear and $b : T \to L_i$. On a supplied binary tree of factors, a subtree S has contribution spaces

$$U_S = \sum_{i \in S} \operatorname{im} a_i, \qquad V_S = \sum_{i \notin S} \operatorname{im} a_i, \qquad w_S = \dim(U_S \cap V_S).$$

Proposition 5.3 (Boundary-coordinate compilation)**.** *For a fixed linear output map $c : T \to W$, $\dim T = t$, feasible partial syndromes at cut S lie in*

$$\operatorname{Hom}(T, U_S) \cap \big(c - \operatorname{Hom}(T, V_S)\big).$$

This set is empty or an affine space with q^{tw_S} elements. Given local choice tables on n disjoint blocks and a binary tree with $w = \max_S w_S$, scalar additive optimization uses at most $O(nq^{2tw})$ pair combinations, in addition to scanning the local tables and polynomial linear algebra. For a finite optimum it returns an attaining lift when local lifts are supplied. A separate integer support budget r increases the bound to $O(n(r+1)^2 q^{2tw})$ combinations.

Proof. If the displayed affine set is nonempty, subtract any one of its elements. The difference space is $\operatorname{Hom}(T, U_S) \cap \operatorname{Hom}(T, V_S) = \operatorname{Hom}(T, U_S \cap V_S)$, of dimension tw_S. Choose affine boundary coordinates by linear algebra. These states express linear compatibility, not necessarily realization by the local choice tables. Store at each state the least cost of actual subtree choices with that contribution, with value ∞ when none exist. Every global feasible assignment survives the restriction, since its complementary choices supply the remaining contribution to c.

At a leaf, scanning its table establishes this invariant. At an internal node, enumerate child-state pairs, add their syndrome maps, discard sums outside the parent affine set, and minimize the summed costs. Disjoint children combine independently. Conversely, a subtree realization at an admitted parent state has admitted child states: the sibling contribution and a linear outside completion supply each child's complement. This preserves the invariant. At the root, return the entry at c, or ∞ if its affine set is empty. The invariant gives the global optimum and detects infeasibility. Each child has at most q^{tw} states, giving the pair bound. Store minimizers and backtrack for finite optima. For a support budget retain the best price at each exact helper count $0, \ldots, r$, starting with the leaf tables, and add counts when combining; disjoint ownership makes those counts additive. ☐

The bound counts state-pair combinations and charges local compilation separately. Coordinate maps are precomputed by polynomial linear algebra; processing a pair has polynomial arithmetic cost in the interface dimensions, in addition to cost addition and comparison. Exact rational prices also incur their arithmetic bit cost. Paired normalization labels must be included when measuring width for a target-normalized query. Support or Pareto messages multiply combination work by the numbers of retained alternatives and their pruning costs. On a balanced tree, a change to one leaf's costs or availability, with the linear interfaces and fixed query unchanged, recomputes only its ancestor messages: $O(C_{\text{combine}} \log n)$ work plus the leaf update. Changing the interface spaces can require new coordinates. None of these bounds avoids the output cost of explicitly requesting all $q^{t \dim W}$ syndromes.

Optimality certificates. An independent certificate for this tree recursion assigns a potential $p_S(s)$ to each subtree state. At a leaf, require $p_S(s)$ to be no greater than the cost of every local choice realizing s. At each internal node with children S_L, S_R, require

$$p_S(s) \leq p_{S_L}(s_L) + p_{S_R}(s_R)$$

for every legal child-state pair with parent state s. Induction bounds the cost of every feasible subtree realization below by its potential. A feasible root witness of cost $p_E(c)$ therefore proves optimality. In the budgeted version the states also include exact helper counts. This certificate requires complete local-choice and transition coverage and justified state restrictions; a reconstructed witness alone supplies only an upper bound. The sequential recursion (17) instead has the usual path certificate, obtained by telescoping edge-potential inequalities on states (i, c).

An economical connected family. Let $W = \mathbb{F}_q^{n+1}$ with basis $e_1, \dots, e_{n+1}$, and give block i three disjoint physical helpers with columns $(e_i, e_{i+1}, e_i + e_{i+1})$. These store two adjacent message symbols and their sum; adjacent blocks share a message symbol but no physical helper. The query can be any fixed map $c : T \to W$, including, at rank one, the sum of all $n+1$ message symbols. Each local coefficient map has $3t$ entries, so enumerating q^{3t} maps constructs its entire labelled cost table, with witnesses, in polynomial arithmetic work per map.

Use a balanced binary tree of contiguous intervals of blocks. For an interval $S = [i, j]$, $U_S = \langle e_i, \dots, e_{j+1} \rangle$. Its intersection with the outside contribution space is spanned by at most the two endpoint vectors e_i, e_{j+1}; an endpoint is absent when that side has no outside block. Thus $w \leq 2$, and every boundary coordinate is an endpoint coefficient. No search for a decomposition or implicit interface is needed. The local-table scans cost $O(nq^{3t})$ candidate maps, and the tree costs $O(nq^{4t})$ state-pair combinations, with polynomial arithmetic overhead. By comparison, the direct ambient-state enumeration in Proposition 5.1 has $Q = q^{(n+1)t}$ and $S = q^{2t}$, hence $O(nq^{(n+3)t})$ transitions. For fixed q, t, this is an explicit family with linear table and combination counts instead of exponentially many ambient states. The comparison is to that enumeration, not a lower bound for other algorithms.

5.5 A worked bilinear task reduction

Let worker i return $a_i(x)b_i(y)$, with $a_i \in V^*$ and $b_i \in W^*$ over $\mathbb{F}_q$. For $\tau_i = a_i \otimes b_i$, define $Q : \mathbb{F}_q^E \to V^* \otimes W^*$ by $Qe_i = \tau_i$. A bilinear target f has a linear aggregation on workers H exactly when $f = \sum_{i \in H} c_i \tau_i$: evaluate on pairs of basis vectors to prove necessity, and use bilinearity for sufficiency. Taking $D = Q^{-1}(\langle f \rangle)$ and $K = \ker Q$ makes this a prescribed quotient lift when $f \in \operatorname{im} Q$.

For example, three workers return

$$x_1y_1, \qquad x_2y_2, \qquad (x_1+x_2)(y_1+y_2),$$

and the target is $x_1y_2 + x_2y_1$. Comparing the four tensor coefficients forces the aggregation coefficients $(-1,-1,1)$ over every field, including characteristic two. All three workers are necessary, the contacted-worker cost is three, and the nonnegative weighted cost is $p_1 + p_2 + p_3$. If any worker is unavailable the target is outside the available span. Generally, a separating linear functional that vanishes on that span but not on f certifies this impossibility, by extending a basis of the span. It certifies impossibility of the stated linear aggregation protocol.

Compression to a smaller worker-function space is exact when a linear map is injective on the span of the workers and requested tasks: every coefficient relation then holds before compression exactly when it holds afterward. This preserves supports, prices, and aggregation witnesses. A Schur-product presentation can be used when it supplies this identification. For repeated inputs or polynomial tasks, the space must be the actual function space; formal monomials need not be independent over a finite field.

5.6 From a compiled choice to a repair decision

For a single request, Ergodis returns the least helper cost and recovery coefficients (`DESIGN.md`, `src/composition.rs`); backtracking expands compatible intermediate labels to physical helper lifts.

For a workload, Ergodis retains compressed support families and resource-load alternatives to select repairs that can coexist. Pareto search and Lagrangian bounds use those alternatives to compare assignments under shared capacities (`OPTIMIZATION.md`, `src/scheduler.rs`, and `src/scheduler_bound.rs`). The price formulation above explains a further integration: compositional pricing can supply a missing repair together with its coefficients, and exact pricing can establish that the fractional scheduler has omitted no improving option. This implication requires an oracle for the full admitted family; supplying a list of options to a scheduler alone does not establish it.

Failures restrict choices and congestion changes their valuation; the ancestor-update bound localizes reevaluation on a fixed tree. Ergodis also implements finite observational minimization and checks whether an additional readout is constant on the retained classes (`src/contextual.rs`). This makes the choice of observable explicit: a quotient sufficient for an old numerical query may need refinement before it can answer a new price or availability query.

The named files and the composition checker `crates/verify/src/binary_composition.rs` have content identities in `verification/artifact-versions.json`. Section 10 states the checking scope and formal coverage; these source references do not record a fresh execution audit.

5.7 Relation to adjacent formulations

The main results specialize established support, duality, and elimination principles to recovery requests. The table separates those inputs from the additional conclusions proved here.

Result	Established input	Recovery conclusion
Theorems 3.3, 4.1	Relative weights and prescribed-tuple coset support	The canonical target/helper pair gives local costs; compatible outer labels and the nonzero inner-kernel cost give exact equation escape.
Theorem 4.2	Dual-word witnesses and component recovery transport	Target-touching distance forces every bounded minimal global support to be local, even when equations can escape.
Theorem 4.4	Functional-dual decomposition and min–sum elimination	Target normalization and functional labels compose with attaining coefficient lifts.
Proposition 5.3	Linear cut-state compression	A fixed recovery query uses affine boundary coordinates; local compilation and output costs are charged separately.
Theorem A.3	Indistinguishability under exterior contexts	Helper radius and target rank bound the length and dual dimension of separating recovery contexts.

Support and recovery. Relative generalized Hamming weights and relative dimension/length profiles already describe support thresholds for nested codes [27, 24, 17, 38, 16]. Recovery-set formulations retain the choices needed for availability and allocation [18, 35, 37, 1, 19]. Coefficients are also established: Márquez-Corbella, Martínez-Moro, and Munuera characterize recovery by dual words and minimal recovery by minimal dual words, and give executable recovery methods [30, Proposition 1 and Corollary 2]. Galindo et al. transport a component recovery set and its coefficients to a matrix-product code [15, Proposition 8]; concatenation also yields LRC constructions [25, 22]. Our reverse-control question asks when every bounded global equation, or every bounded minimal support, must come from the target block.

Service-rate regions allocate requests under helper capacities [3, 4]; inclusion-minimal supports suffice [5, Proposition 3.3]. Our operational transfer preserves that bounded family. Disjoint simplex repair [8], online batch recovery [11], and download/subpacketization models [39, 43, 21, 31] pose distinct workload or resource constraints.

Composition and state. Zhang et al. define minimum union support for a prescribed tuple of cosets [44]; maximizing over tuples gives generalized covering quantities [12]. Functional-dual decomposition and min–sum elimination occur in concatenated decoding [14, 9, 20, 6]. The generalized distributive law supplies the general message-passing principle [2, Theorem 3.1]. The recovery formula specifies its fibres, normalization, and coefficient lifts; it does not introduce min–sum elimination. Labelled soft information and presentation-dependent enumerators serve related decoding and enumeration tasks [26, 34].

Kashyap's linear cut-state quotient provides the coding-theoretic setting for boundary compression [23, Section 2.4, equation (1), and Theorem 3.4]. Contextual equivalence is likewise established in weighted automata and tree series [33, Section 6.4, Theorem 8], [29]. Funk, Mayhew, and Newman define indistinguishability by exterior extensions and prove its union congruence [13, Definition 2.2 and Proposition 2.3]. Appendix A fixes the original leaf confinement boundary and derives concrete recovery probes and a radius-controlled separator bound. Its numerical observations do not include arbitrary reweightings or escape from newly formed macroblocks.

6 Sharp limits of coarser recovery invariants

The preceding sections identify exact information for three different tasks. Here we show that two tempting coarsenings lose essential information. The first separation rules out the complete RGHW hierarchy as a reliability invariant at every quotient rank: minimum support sizes do not retain the overlap pattern of recovery sets. The second rules out even the full abstract nested pair, together with the same unlabelled helper supports, as a confinement invariant: its ambient presentation can change $d(I^{\perp})$ and the prescribed-coset costs in (9).

First consider a one-dimensional recovered space. On the helper set $\{0, \dots, 8\}$, let

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\},$$

where, for example, 158 denotes $\{1, 5, 8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10, 4, 6]_q$ code over a common finite prime field. These are the complete radius-three families of inclusion-minimal recovery supports. Larger minimal supports also occur: in $\mathcal{A}$, $\{0, 1, 2, 4\}$ contains none of the listed triples, while the construction below makes its four helper columns a basis and hence a recovery set.

Proposition 6.1 (Rank-one support-overlap separation)**.** *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s, their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \tag{19}$$
$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \tag{20}$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability. Indeed, $R_{\mathcal{A}}(s) - R_{\mathcal{B}}(s) = s^6(1 - s)^3$, so the first family has strictly larger radius-three reliability for every $0 < s < 1$ under this homogeneous independent-survival law.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \dots, 5$, the numbers of k-edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9.

Inclusion–exclusion for the five edge-containment events gives (19) and (20).

Realize the listed triples as plane collinearities, then lift so that precisely their unions with the target x are four-column circuits.

The two plane arrangements. Work over $K = \mathbb{Q}$. In homogeneous coordinates (X, Y, Z), choose five projective lines $\ell_1, \dots, \ell_5$ using the following defining linear forms:

	ℓ_1	ℓ_2	ℓ_3	ℓ_4	ℓ_5
$\mathcal{A}$	Z	$X - Y$	Y	X	$X + Y + Z$
$\mathcal{B}$	X	$X + Y + Z$	Z	Y	$X - Y$.

For $\mathcal{A}$, the lines ℓ_2, ℓ_3, ℓ_4 are concurrent; label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose $2, 6$ on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, the lines ℓ_1, ℓ_4, ℓ_5 meet at 1; put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. The fixed points in these two arrangements have representatives

$$\begin{aligned} \mathcal{A}: &\quad 0 = e_3, \quad 1 = e_2, \quad 4 = e_1 - e_3, \quad 5 = e_1, \quad 8 = e_1 - e_2; \\ \mathcal{B}: &\quad 1 = e_3, \quad 8 = e_2, \quad 7 = e_1 - e_2, \quad 0 = e_1, \quad 2 = e_1 + e_2 - 2e_3. \end{aligned}$$

The only collinear triples among these fixed points are $045, 158$ in $\mathcal{A}$ and 078 in $\mathcal{B}$, as their coordinates show. Choose each remaining point on its assigned line, avoiding previously chosen points, intersections with other supporting lines, and unintended joining lines of pairs already chosen. No unintended joining line coincides with the assigned line: its existing points belong to the prescribed triple being completed. Thus only finitely many rational points are excluded at each step. The five displayed triples are exactly the resulting collinear triples.

Generic lift. Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i. For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i. No four helper points are collinear, since every supporting line contains exactly three selected points and every other collinear triple was excluded. Their quotient vectors therefore span K^3. Expanding the lifted determinant along the first coordinates exhibits a nonzero cofactor, so it is a nonzero linear form in those coordinates. Avoiding this finite union of proper hyperplanes makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A.

Finite-field realization and parameters. Clear the column denominators in both rational configurations. For every subset required to be independent, retain a nonzero integer minor witnessing that independence, including triples containing x and two helpers. Such triples are independent because the projected helper points are distinct. Choose one prime dividing none of these finitely many minors. The required dependence determinants are zero and remain zero on reduction. Thus both row-span codes have rank four, dual distance four, and minimum-cardinality recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. □

Concrete realizations. Over $\mathbb{F}_{101}$, the following generator matrices realize the two families, with columns ordered as target x, then helpers $0, \ldots, 8$:

$$G_{\mathcal{A}} = \begin{pmatrix} 1 & 66 & 83 & 80 & 96 & 37 & 73 & 9 & 34 & 92 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 2 & 100 \\ 0 & 1 & 0 & 1 & 4 & 100 & 0 & 2 & 98 & 0 \end{pmatrix},$$

$$G_{\mathcal{B}} = \begin{pmatrix} 1 & 72 & 33 & 20 & 51 & 31 & 4 & 8 & 61 & 20 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 & 0 & 1 & 100 & 1 \\ 0 & 0 & 1 & 99 & 98 & 5 & 2 & 1 & 0 & 0 \end{pmatrix}.$$

All three-column sets and all helper four-column sets are independent; the dependent four-sets through x are precisely the five specified circuits. The accompanying explicit-example certificate checks these statements by elimination and independent determinant expansion, and checks the reliability polynomials by enumerating availability sets. These instances illustrate the preceding general existence proof.

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius-R full-quotient reliability by

$$\Pr\{\exists L \leq D : \ L + K = D, \ |\operatorname{supp} L| \leq R, \ \operatorname{supp} L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s. The condition $L + K = D$ says that the surviving helper equations recover the entire quotient D/K; the radius R bounds their union support $|\operatorname{supp} L|$.

Theorem 6.2 (The complete RGHW hierarchy does not determine reliability)**.** *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \ldots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 6.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \qquad (1 \leq i \leq \ell - 1),$$

where each padding line has the unique support of size L_i. Thus

$$K = K_0 \oplus 0 \oplus \cdots \oplus 0, \qquad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \cdots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the tth relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \ldots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t-dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R. Thus

$$R_{\mathrm{full},R}(s) = s^{\sum_i L_i} R_{\mathrm{base},3}(s).$$

The two polynomials remain different by (19)–(20).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J/K$, let $\pi : \mathbb{F}_q^J \to V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \to D/K \le V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \qquad G(a,x) = \alpha(a) + \pi(x),$$

uses the first ℓ coordinates as targets, J as helpers, and has message space V. Its helper kernel is K and the helper preimage of the target span is D, so its associated nested pair is $K \subseteq D$. $\square$

The second separation fixes the abstract nested pair itself. This is not a row-basis change of one inner code: by Proposition 3.2, such a change leaves (D_P, K_P) unchanged. Here a *coefficient presentation* means an ambient inner dual code, target coordinates, and target coefficient map that realize a prescribed abstract pair $K \subseteq D$ as its shortening–puncturing pair. Different such realizations may have different ambient dual codes and distances, so the examples below cannot be related by changing the row basis of one encoder. Let

$$K = 0, \qquad D = \langle u, v\rangle \le \mathbb{F}_2^3, \qquad u = 100, \quad v = 011.$$

The nonzero helper words have weights $1, 2, 3$, so $(M_1, M_2) = (1, 3)$.

Proposition 6.3 (The abstract nested pair does not determine confinement)**.** *Present the target space first by*

$$10 \mapsto u, \qquad 01 \mapsto v, \qquad 11 \mapsto u+v,$$

and then by

$$11 \mapsto u, \qquad 10 \mapsto v, \qquad 01 \mapsto u+v.$$

The two inner codes have the same associated nested pair, the same unlabelled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2,3,5\}$ *and* $\{3,3,4\}$*. In every outer-distance regime where the collapse theorem applies, their additive costs are respectively*

$$(M_1 + d(I^\perp), M_2 + d(I^\perp)) = (3,5) \qquad \textit{and} \qquad (M_1 + d(I^\perp), M_2 + d(I^\perp)) = (4,6).$$

Proof. Use the graph-code realization

$$I^\perp = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification from the two-dimensional target space to D. Over $\mathbb{F}_2$ the minus sign is invisible. The three nonzero graph words are

	target coefficient	helper word	target/helper weights	total weight
I_1	10	100	$1+1$	2
	01	011	$1+2$	3
	11	111	$2+3$	5
I_2	11	100	$2+1$	3
	10	011	$1+2$	3
	01	111	$1+3$	4

Thus $d(I^\perp)$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1,3)$ gives the two threshold sequences by Theorem 7.2. $\square$

These are limits on representation, not on executability. Keeping the functional labels avoids the second loss, and keeping minimizing lifts permits the exact recursion to return the original coefficient systems. Section 5 implements this closure law as an exact recursion.

7 Distance specializations and bounded transfer

The exact optimization also recovers the outer-distance collapse without choosing a radius. Indeed, every nonzero functional sector costs at least $d(O^{\perp}) - 1$: its label has at least $d(O^{\perp})$ active blocks, the target block may contribute zero helpers, and every other active block contributes at least one. The zero-functional sector attains $M_t(D_P, K_P) + d(I^{\perp})$ after minimization over T. Consequently,

$$d(O^{\perp}) \geq M_t(D_P, K_P) + d(I^{\perp}) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^{\perp}). \tag{21}$$

Theorem 7.1 (Confinement for a fixed target subspace)**.** *Let r be a nonnegative integer and let $O \leq L^N$ be an L-linear outer code with $N \geq 2$ and $d(O^{\perp}) > r + 1$, and assume $0 \neq T \leq W_P$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^{\perp}). \tag{22}$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r.

Proof. If $0 \neq B \in \mathcal{B}_T(O)$, choose $u \in T$ with $B(u) \neq 0$. The tuple $B(u) \in \mathrm{FD}(O)$ has at least $d(O^{\perp})$ nonzero blocks. At most one is the target block, so every lift has at least $d(O^{\perp}) - 1 > r$ helpers outside it. Thus no nonzero-functional sector in (9) occurs at cost at most r.

The zero-functional sector has exact cost $\mu_{I,P,T}(0) + d(I^{\perp}) = \rho_T(I) + d(I^{\perp})$. Theorem 4.1 now gives both directions of (22) and the support-preserving bijection. □

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 7.2 (Confinement by recovered dimension under an outer-distance condition)**.** *Let r be a nonnegative integer, let $O \leq L^N$ be L-linear with $N \geq 2$ and $d(O^{\perp}) > r + 1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost-$\leq r$ recovery system for every helper-recoverable t-dimensional target subspace if and only if*

$$r < M_t(D_P, K_P) + d(I^{\perp}). \tag{23}$$

When (23) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r.

Proof. By Theorem 3.3, every such target subspace has inner cost at least $M_t(D_P, K_P)$, and some target subspace attains this minimum. Apply Theorem 7.1 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. □

7.1 One-coordinate specialization

For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \mathrm{wt}(y), \qquad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \mathrm{wt}(y),$$

with value ∞ when a constrained fibre is empty. The first quantity is the usual coset-leader weight for the inner encoder, while the second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [9, Theorem 2.1]. Here a dual word "through x" simply means that its coefficient at x is nonzero. The quantity below counts total dual-word weight; the general recovery cost excludes the forced target coordinate, so for a nonzero target column the two conventions satisfy $Z_{j,x} = \Gamma_{j,T} + 1$.

Theorem 7.3 (Exact weighted finite-length confinement at one coordinate)**.** *Fix an outer block j, an inner coordinate x, and $N \geq 2$, and assume that the coordinate projection $O \to L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j, x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O, I) = \min \left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h)\in \mathrm{FD}(O) \\ \beta \neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h \neq j} \lambda_I(\beta_h) \right) \right\}. \tag{24}$$

A minimum over an empty set is ∞. Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j, x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O, I) > r + 1. \tag{25}$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks and apply (5). For a fixed nonzero functional tuple, the block representatives minimize independently, giving the second term of (24). For the zero tuple, nonconfinement costs $\mu_{I,x}(0)$ in the target block and $d(I^\perp)$ in another block. The projection hypothesis makes confined words exactly the zero-extensions of inner-dual words, so these two sectors prove (24).

If the target column is nonzero, take $P = \{x\}$ and let $T \leq U_P$ be its span. After choosing a nonzero vector of T, a linear map $T \to L^*$ is one functional. Normalizing its target coefficient and rescaling the functional tuple identify the minima in (9) with those in (24). The target coordinate contributes one to total word weight and is not a helper, so

$$Z_{j,x}(O, I) = \Gamma_{j,T}(O, I) + 1.$$

Theorem 4.1 gives the claimed condition. If the target column is zero, the same condition follows directly from the exact minimum already proved. In either case, below the minimum every equation is confined, so restriction and zero-extension are inverse. In one dimension every exact helper support is the support of its normalized equation; the equivalence also holds for exact helper-support families. It implies equality of their inclusion-minimal members, but minimal-support equality need not imply equation confinement, as Theorem 4.2 shows. ∎

Theorem 7.3 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code. Its construction chooses an outer dual repetition line whose projective multipliers avoid every ratio between low-cost inner labels. Consequently the labelled criterion holds even though ordinary distance sees no margin.

Proposition 7.4 (A family from a Singer cycle without the outer-distance condition)**.** *Let $k \geq 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k+1)^2. \tag{26}$$

Let I be the $[k+1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k+1 \leq n \leq 2k+1$, there is an L-linear length-n outer code O with $d(O^\perp) = n$ such that (25) holds at helper radius $n-1$ for every inner coordinate and every outer block. Thus all normalized radius-$(n-1)$ recovery equations and their exact helper supports are copied blockwise although the usual outer-distance condition fails: here $d(O^\perp) = n$ is not strictly greater than $r + 1 = n$.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k+1$ coordinate functionals of the encoder give a set S of $k+1$ points in the projective space on L^*. The multiplication maps of L, modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k-1)/(q-1)$, acting regularly on its points [40]. Indeed, under the trace identification, the multiplier class sending a nonzero functional class $[f]$ to $[g]$ is uniquely $[g/f]$. Each ordered pair of points of S therefore forbids at most one cycle element, so at most $|S|^2 = (k+1)^2$ elements send a point of S to a point of S. Thus (26) permits a multiplication map $a : L \to L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^\times$, whose dual action satisfies $a^*(S) \cap S = \varnothing$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \ldots, f)$. If $f \neq 0$, all n entries are nonzero, so $d(O^\perp) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k+1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S, contrary to the choice of a. Every nonzero tuple therefore costs at least $n+1$.

Every outer coordinate projection is surjective. Finally, $I^\perp$ is spanned by $(1, \ldots, 1, -1)$, so $d(I^\perp) = \mu_{I,x}(0) = k+1$ for every x. The zero-functional cost in (24) is $2k+2 > n$, and the nonzero-functional cost is at least $n+1$. Thus $Z_{j,x}(O, I) > n$ for every j, x. At $r = n-1$ the ordinary distance condition would require $d(O^\perp) > n$, which is false, while the exact weighted condition holds. Taking $(q, k, n) = (5, 3, 5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n - 1 \geq k$, the asserted transfer is nonvacuous throughout the family. $\square$

Explicit field and multiplier. For $(q, k, n) = (5, 3, 5)$, take $L = \mathbb{F}_5[\alpha]/(\alpha^3 + \alpha + 1)$ in the basis $(1, \alpha, \alpha^2)$. The cubic has no root in $\mathbb{F}_5$, so it is irreducible. Multiplication by $\gamma = \alpha + \alpha^2$ has matrix

$$M_\gamma = \begin{pmatrix} 0 & 4 & 4 \\ 1 & 4 & 3 \\ 1 & 1 & 4 \end{pmatrix}.$$

The coordinate functionals of $G = (I_3 \mid \mathbf{1})$ are the rows $e_1^*, e_2^*, e_3^*, (1,1,1)$. Their pullbacks by M_γ are

$$(0,4,4), \quad (1,4,3), \quad (1,1,4), \quad (2,4,1).$$

None is projectively a coordinate row or $(1,1,1)$, proving the required disjointness directly. The outer code is $O = \{z \in L^5 : z_0 + \gamma z_1 + z_2 + z_3 + z_4 = 0\}$. This specifies the encoder alignment as well as the field and parameters.

For a nonzero, helper-recoverable singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x. If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp). \tag{27}$$

The helper-radius condition (23) is therefore $r + 1 < z_x(I)$, with no change of convention hidden in the reduction. If the target column is nonzero but not helper-recoverable, then $W_P = 0$: its local recovery cost is infinite and there is no first relative weight of this pair. Theorem 4.1 is the common finite formula. The outer-distance condition removes its nonzero-functional sector and gives the RGHW threshold in every recovered dimension. Restricting instead to one target coordinate gives Theorem 7.3 without that condition.

The labelled functions suffice to compose fibre minima. Computing nonconfinement also retains the nonzero-kernel cost $d(I^\perp)$ through (15); its ordinary zero-label minimum is zero and does not encode this cost. Minimizing lifts must also be stored to propagate witnesses. Coarser summaries answer different questions. Section 6 shows that relative-weight minima do not determine recovery reliability and that the abstract nested pair does not determine confinement.

8 Consequences of the threshold

The fixed-target cost refines ordinary generalized weights, which optimize over target sets, and is bounded above by worst-case cooperative locality, which maximizes over them. Let $d_e(C^\perp)$ denote the eth generalized Hamming weight of $C^\perp$ [42]. For an e-set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e-dimensional subcode $A \leq C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$, so every target coefficient vector has a unique equation in A. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

The first identity below is the standard information-set interpretation of generalized Hamming weights; we record it to derive the paper-specific nonconfinement consequence.

Theorem 8.1 (Best-target generalized weight and first nonconfinement cost)**.** *For $1 \leq e \leq \dim C^\perp$,*

$$\min_{\substack{P \subseteq E \\ |P| = e}} \kappa_C(P) = d_e(C^\perp) - e. \tag{28}$$

Consequently, if every e-set is recoverable and $r_e^{\mathrm{coop}}(C) = \max_{|P|=e} \kappa_C(P)$, then

$$r_e^{\mathrm{coop}}(C) \geq d_e(C^\perp) - e.$$

Represent C by an encoder with message alphabet L and let $O \leq L^N$ be L-linear, with $N \geq 2$. Put $a = d_e(C^\perp) - e$. If

$$d(O^\perp) \geq a + d(C^\perp) + 1,$$

then the least helper cost of a nonconfined identity-normalized system in $O \circ C$, minimized over all e-coordinate target sets in a chosen block, is

$$d_e(C^\perp) - e + d(C^\perp). \tag{29}$$

Proof. A normalized identity system on P spans an e-dimensional subcode of $C^\perp$ whose support is P together with its helper union. Hence $d_e(C^\perp) \le e + \kappa_C(P)$.

Conversely, let $A \le C^\perp$ have dimension e and support size $d_e(C^\perp)$. Choose e linearly independent columns of a generator matrix of A; their coordinate set P is an information set and has size e, and restriction $A \to \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\operatorname{supp}(A) \setminus P$. This proves (28); the cooperative-locality inequality follows by taking the maximum instead of the minimum; compare [1, Theorem 1].

It remains to justify the nonconfinement formula when the target columns indexed by P are dependent. Put $U = \operatorname{im} G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \to \mathbb{F}_q^P$ gives a system on U. Conversely, given a system on U, write

$$a = \alpha(G_P a) + \big(a - \alpha(G_P a)\big) \qquad (a \in \mathbb{F}_q^P).$$

Thus $G_P : \mathbb{F}_q^P \twoheadrightarrow U$ separates each target vector into its chosen lift $\alpha(G_P a)$ and a kernel relation. The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

The distance hypothesis implies that every outer coordinate projection is surjective. Apply the block-functional decomposition to an identity system parameterized by $\mathbb{F}_q^P$. In the zero-functional sector its target block uses at least $\kappa_C(P)$ helpers. Nonconfinement requires a nonzero map into the inner dual in another block, costing at least $d(C^\perp)$. Conversely, when P is recoverable, attach a minimum-weight dual word in a second block through a nonzero functional on $\mathbb{F}_q^P$. Thus the zero-sector cost is exactly $\kappa_C(P) + d(C^\perp)$, with value ∞ when P is not recoverable. This argument also covers $\operatorname{im} G_P = 0$, since the identity parameter space still has dimension $e \ge 1$.

Every nonzero-functional sector uses at least $d(O^\perp) - 1$ helpers, hence at least $a + d(C^\perp)$. Every zero-sector cost has the same lower bound, and an information-set target attaining (28) attains it. This proves (29). Equality in the distance bound can permit a nonzero-sector tie; exclusion of that sector is unnecessary. □

This hypothesis controls the minimum over target sets. For an individual recoverable P, the sufficient bound is $d(O^\perp) \ge \kappa_C(P) + d(C^\perp) + 1$. If every e-set is recoverable, replacing $\kappa_C(P)$ by $r_e^{\mathrm{coop}}(C)$ forces these individual collapse formulas simultaneously.

Before bounding the MDS thresholds, we record what the coefficient layer adds even when the support clutter is completely uniform.

Theorem 8.2 (Minimum-weight MDS recovery equations span the dual)**.** *Let $C \le \mathbb{F}_q^E$ be an $[n, k]_q$ MDS code with $1 \le k < n$, and let $x \in E$. Then*

$$\operatorname{span} \mathcal{P}_x^{(\le k)}(C) = C^\perp,$$

and their supports form the complete k-uniform clutter on $E \setminus \{x\}$.

Equivalently, the minimum-weight dual equations through x realize every k-subset of the other coordinates as a helper support.

Proof. The dual has dimension $n-k$ and distance $k+1$. For each $(k+1)$-set T, restriction $C^{\perp} \to \mathbb{F}_q^{E\setminus T}$ has a nonzero kernel, whose words have support exactly T by the distance bound. Thus all stated supports occur. Fix a $(k-1)$-set $Q \subseteq E \setminus \{x\}$ and take the normalized word on $\{x,t\}\cup Q$ for each $t \in E\setminus(\{x\}\cup Q)$. Their private coordinates t make these $n-k$ words independent, hence a basis of $C^{\perp}$. □

The support clutter depends only on the MDS parameters, whereas the normalized equations retain the represented dual code.

Put $k = \dim I$, $u = \operatorname{rank} G_P$, and $b = \operatorname{rank} G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \le b - \ell + t, \qquad d(I^{\perp}) \le k+1. \tag{30}$$

The three ranks governing the statement are

$$\begin{array}{r|l} u = \operatorname{rank} G_P & \text{dimension visible on the target coordinates} \\ b = \operatorname{rank} G_J & \text{dimension visible on the helper coordinates} \\ \ell = \dim(\operatorname{im} G_P \cap \operatorname{im} G_J) & \text{dimension recoverable from the helpers.} \end{array}$$

Theorem 8.3 (MDS ceiling and rank-one rigidity)**.** Universal ceiling. *For* $1 \le t \le \ell$*,*

$$M_t(D_P, K_P) + d(I^{\perp}) \le 2k - \ell + t + 1. \tag{31}$$

MDS staircase. *Suppose I is MDS. Then*

$$u = \min\{k, |P|\}, \qquad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \ge 1$, then, for $1 \le t \le \ell$,

$$M_t(D_P, K_P) = k - u + t, \qquad M_t(D_P, K_P) + d(I^{\perp}) = 2k - u + t + 1. \tag{32}$$

If $|J| \ge k$, then $\ell = u$ and these values attain the ceiling in (31) at every rank. Rank-one rigidity. *Conversely, when $\ell \ge 1$, equality in (31) at $t = 1$ forces I to be MDS and forces equality at every rank.*

Proof. The ceiling follows from (30) and $b \le k$. For an MDS code every set of at most k generator columns is independent; equivalently, the column matroid is uniform. Thus $\operatorname{rank} G_P = u$, $\operatorname{rank} G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \le k$,

$$\dim\bigl(\operatorname{span} G_P \cap \operatorname{span} G_H\bigr) = \max\{0, u + h - k\}.$$

This is the usual dimension count for two subspaces generated by disjoint subsets of a uniform k-dimensional column configuration. Thus Proposition 3.4 gives $M_t = k - u + t$ for $1 \le t \le \ell$; here $k - u + t \le b \le |J|$. Now $d(I^{\perp}) = k+1$, and if $|J| \ge k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \le k$. Hence $d(I^{\perp}) = k+1$; equality in the dual Singleton bound is precisely the MDS condition for I. Also $M_1 = b - \ell + 1$. Strict growth then gives $M_t \ge b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. □

Thus extremality is detected at rank one: equality in the first threshold forces the inner code to be MDS and forces every later rank to attain its ceiling. No separate higher-rank hypothesis is needed.

We next show that the transferred local structure can occupy a positive fraction of an asymptotically good concatenated family, rather than appearing only in one distinguished block. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $\Delta_N = d(O_N)$.

Corollary 8.4 (Positive-density realization)**.** *Assume $d(O_N^\perp) \to \infty$. For every fixed finite radius r, the inclusion-minimal recovery supports for every nonzero $T \le W_P$ are eventually copied in every block, with no inner additive ceiling on r. For the stronger coefficient statement, fix either a nonzero target subspace T and a radius r satisfying Theorem 7.1, or a rank t and radius r satisfying Theorem 7.2. For all sufficiently large N, every corresponding inner normalized recovery system of cost at most r and its exact helper support are copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density $|P|/m$.*

In particular, if $r < M_1(D_P, K_P) + d(I^\perp)$, then all cost-$\le r$ systems for all nonzero subspaces of W_P are copied simultaneously in every block by Corollary 4.5. Hence the complete bounded local recovery data through radius r occur on the stated positive-density coordinate classes.

The concatenated code has length mN, dimension kK_N, and minimum distance at least $d\Delta_N$. If

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{\Delta_N}{N} \ge \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families with positive limiting rate, positive primal relative distance, and dual distance tending to infinity exist over every finite field L.

Proof. Minimal-support transfer follows from Theorem 4.2, since $\delta_j(O_N^\perp) \ge d(O_N^\perp)$ uniformly in j. The coefficient transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N. A nonzero outer word has at least Δ_N nonzero symbols, each encoding to an inner word of weight at least d. This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^\perp$ such that

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R,$$

where H_Q is the standard Q-ary entropy function measuring the exponential growth rate of a Q-ary Hamming ball. The first-moment argument for a random K_N-dimensional L-linear code bounds the expected numbers of nonzero primal and dual words below the two cutoffs, up to subexponential factors, by

$$Q^{N(H_Q(\delta)+R-1)} \qquad \text{and} \qquad Q^{N(H_Q(\delta^\perp)-R)},$$

respectively. Both tend to zero by the displayed inequalities, so one may choose a sequence with rate tending to R, primal relative distance at least δ, and dual relative distance at least $\delta^\perp$; see also [28, Chapter 17]. □

Minimal-support transfer preserves the fractional load-balancing region used in service-rate problems. The target-touching distance condition is sufficient even when coefficient confinement fails.

Fix one inner target block with helper set J. Let $\mathcal{A}$ be a finite set of demands on that block, with demand a represented by a nonzero target subspace $T_a \le W_P$, and let $\mathcal{M}_a^{(\le r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector c, define

$$\Lambda_{\mathcal{A}}^{(\le r)}(c) = \left\{ \lambda \ge 0 : \begin{array}{l} \text{there are } f_{a,R} \ge 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\le r)}} f_{a,R}, \\ \displaystyle\sum_a \sum_{R \ni j} f_{a,R} \le c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [3, 4, 32, 10]. The use of inclusion-minimal sets is justified by the standard flow reassignment argument [5, Proposition 3.3]; the proof below applies it with the radius constraint and arbitrary nonnegative capacities.

Corollary 8.5 (Transfer of bounded service-rate regions)**.** *Let $O \le L^N$ be an outer code with $N \ge 2$ whose projection onto the target block is nonzero. Suppose either*

$$\delta_j(O^\perp) > r + 1, \qquad \textit{or} \qquad r < \Gamma_{j,T_a}(O, I) \quad \textit{for every } a \in \mathcal{A}.$$

Transport capacities from the inner helper set to its copy in that block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality, because every minimal support is local. The same transfer holds for support-based integral allocations, obtained by requiring the variables $f_{a,R}$ to be nonnegative integers. Consequently, for an outer family with $d(O_N^\perp) \to \infty$, both equalities hold eventually at every fixed finite radius, without an inner-distance inequality.

Proof. Theorems 4.2 and 4.1 give minimal-support equality under the respective hypotheses. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. These operations preserve integrality, proving the integral assertion as well. $\square$

The next family realizes all three information levels in closed form: its RGHWs give minimum recovery costs, its ambient dual distance gives the confinement thresholds under the outer-distance condition, and its projective rank distribution gives reliability beyond the RGHW hierarchy.

9 The projective-simplex family

This non-MDS family realizes the paper's information hierarchy in one model. Its relative weights determine the minimum costs, its ambient inner dual gives the collapsed confinement thresholds, and the full projective rank distribution determines the probability that some t-dimensional target space remains recoverable. This allows the target space to depend on the surviving helpers, unlike recovery of a prescribed T. Thus the family shows both what the RGHW hierarchy captures and what it forgets. Assume $m \ge 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\mathrm{PG}(m-1, q)$. Here $\mathrm{PG}(m-1, q)$ is the set of one-dimensional subspaces of $\mathbb{F}_q^m$, and A chooses one nonzero vector from each such subspace. Define the inner code by

$$I_m^\perp = \{(a, aA) : a \in \mathbb{F}_q^m\}. \tag{33}$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q-ary simplex code. The generalized-weight formula for S_m is classical; in fact every t-dimensional simplex subcode has the same support size [36]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 9.1 (Projective-simplex thresholds)**.** *For $1 \leq t \leq m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q-1}, \tag{34}$$

and

$$d(I_m^\perp) = q^{m-1} + 1. \tag{35}$$

Hence

$$C_t := M_t(S_m, 0) + d(I_m^\perp) = \frac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1. \tag{36}$$

In outer families with dual distance tending to infinity, the collapsed threshold C_t equals $\Gamma_{j,t}(O, I_m)$ once the outer dual distance reaches $C_t + 1$. Equivalently, at every fixed helper radius the corresponding confinement statement holds for all sufficiently large members of the family. The family is non-MDS for $m \geq 2$; for $m \geq 3$ the successive increments of C_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^\perp$ is a, and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m; this proves the asserted identification of the associated pair.

Let $U \leq \mathbb{F}_q^m$ have dimension t. The common zero coordinates of the subcode UA are the projective points in $U^\perp$, a subspace of projective dimension $m - t - 1$. Thus

$$|\operatorname{supp}(UA)| = N - \frac{q^{m-t} - 1}{q-1} = \frac{q^m - q^{m-t}}{q-1},$$

independently of U. This proves (34). Every nonzero simplex word aA has weight q^{m-1}, while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (35). The confinement formula is Theorem 7.2. The non-MDS and increment assertions follow from the displayed parameters. ☐

question	retained information	exact answer
minimum rank-t helper union	RGHW hierarchy	$\frac{q^m - q^{m-t}}{q-1}$
eventual first nonconfined cost	RGHW plus $d(I_m^\perp)$	$\frac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1$
rank-t recovery reliability	projective rank distribution	$R_t(s)$ in (38)

Table 1: The projective-simplex family separates the three questions. The first two use minimum-cost data; the stochastic law requires the distribution of ranks of failed projective sets.

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s, and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = \big(\operatorname{span} F\big)^\perp,$$

which has dimension $m - \operatorname{rank}(F)$. Write $\begin{bmatrix} a \\ b \end{bmatrix}_q$ for the Gaussian binomial coefficient, the number of b-dimensional vector subspaces of $\mathbb{F}_q^a$. A projective r-frame here means r linearly independent projective points, or equivalently the projectivization of a vector-space basis of its r-dimensional span.

Theorem 9.2 (Projective rank reliability)**.** Exact probability. *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \mathrm{PG}(m-1,q) \\ \operatorname{rank}(F) \leq m-t}} (1-s)^{|F|} s^{N-|F|}. \tag{37}$$

Closed form. *Writing $N_v = (q^v - 1)/(q-1)$, this equals*

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^{u} \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \tag{38}$$

Endpoint behavior. *Its endpoint terms are*

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \qquad (s \to 0), \tag{39}$$

$$1 - R_t(s) = B_{m,m-t+1}(1-s)^{m-t+1} + O((1-s)^{m-t+2}) \qquad (s \to 1), \tag{40}$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\operatorname{GL}(r,q)|}{r!(q-1)^r}$$

is the number of unordered projective r-frames.

The first endpoint says that rare successful recovery is dominated by the $\begin{bmatrix} m \\ t \end{bmatrix}_q$ minimum helper supports of size M_t. The second says that near-perfect availability first fails on a smallest projective frame whose rank exceeds the allowed failure span.

Proof. Recovery of t independent target combinations is possible exactly when $m - \operatorname{rank}(F) \geq t$, which gives (37). For a vector subspace $U \leq \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice [41, Proposition 3.7.1 and Example 3.10.2] gives the total probability of failed sets spanning a fixed u-space U as

$$\sum_{V \leq U} \mu(V,U) s^{N-N_{\dim V}}, \qquad \mu(V,U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \qquad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\begin{bmatrix} m \\ u \end{bmatrix}_q$ choices for U and $\begin{bmatrix} u \\ v \end{bmatrix}_q$ subspaces $V \leq U$ of dimension v. Summing over $u \leq m - t$ proves (38).

For $s \to 0$, a smallest surviving set that permits rank-t recovery is the complement of the points in an $(m-t)$-dimensional vector subspace. It has size M_t, and there are $\begin{bmatrix} m \\ m-t \end{bmatrix}_q = \begin{bmatrix} m \\ t \end{bmatrix}_q$ such subspaces. For $s \to 1$, failure first occurs when the failed points span dimension $m - t + 1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. ∎

For an m-dimensional helper code $D \le \mathbb{F}_q^N$ with lower code $K = 0$ and $N = (q^m - 1)/(q - 1)$, the equality $M_1(D, 0) = q^{m-1}$ characterizes the projective-simplex column multiset up to scaling and permutation. Averaging weights over the nonzero codewords gives at most $Nq^{m-1}(q-1)/(q^m - 1) = q^{m-1}$, with strict inequality if a column is zero. Thus equality of the minimum forces every column to be nonzero and every nonzero word to have weight q^{m-1}. For $m \ge 2$, every projective hyperplane consequently contains the same column multiplicity. Summing these multiplicities over the hyperplanes through a fixed point shows that its multiplicity is constant: that point lies in $(q^{m-1} - 1)/(q - 1)$ such hyperplanes and each other point in $(q^{m-2} - 1)/(q - 1)$. Their difference is q^{m-2}. The total multiplicity is N, so each of the N points occurs once. The case $m = 1$ is immediate. This is also the multiplicity-one case of Bonisoli's classification of linear equidistant codes [7].

10 Verification scope

Every theorem has a human proof independent of software execution. The displayed finite instances additionally have the arithmetic checks documented in `verification/explicit-examples.md`. The paper-owned Lean companion additionally verifies the associated-pair exact sequence

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0.$$

Its formal coverage stops there: the relative-weight identity, confinement and composition theorems, operational consequences, and examples are not established by that companion. The claim manifest records this distinction statement by statement. The four checked declarations, pinned toolchain, Mathlib revision, and axiom inventory are documented in the companion's `lean/README.md` and `verification/README.md`.

Ergodis's independent checkers validate specified finite composition and transition contracts. Relating a source recovery problem to such a checked presentation requires justification of the lowering, local costs, and state reductions; checking a reconstructed witness alone proves feasibility. The source versions and historical benchmark records are identified by `verification/artifact-versions.json`. Detailed measurements and replay protocols belong to the software documentation at `tavisrudd/ergodis`. No benchmark or exhaustive computation is a premise of a manuscript theorem; the finite reliability table is evaluated directly by inclusion–exclusion.

11 Conclusion

Functional labels make local recovery costs composable: fixing the intermediate operations separates the block fibres, and retaining their lifts reconstructs the recovery coefficients. The resulting exact equation threshold includes external dual perturbations that create no useful repair alternative. Thus the equation threshold $\Gamma_{j,T}$ and the first new external repair threshold $\varepsilon_{j,T}$ answer different questions. Target-touching outer dual distance gives a sufficient condition for all bounded minimal supports to remain local; it need not determine $\varepsilon_{j,T}$. Reliability and capacity-constrained allocations then transfer without the additional inner-dual distance restriction.

The downstream question determines what the optimizer must retain. Numerical costs suffice for their declared contexts; support alternatives preserve reweighting, availability, and helper sharing. For adjacent-pair helper blocks, local tables and endpoint interfaces can both be constructed with work linear in the number of blocks at fixed field and target rank. The remaining algorithmic question is which broader recovery families admit such economical interface construction, including local compilation and certification.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

A Contextual equivalence and rank reduction

The labelled composition law suffices for exact recovery optimization. This appendix asks which numerical distinctions can be removed when only escape costs at a fixed original leaf boundary are observed. The resulting finite tests do not replace support families for availability or scheduling.

We first record a normal form for outer functional duals. It will let us replace an arbitrary outer code by the finite interface that a bounded-cost recovery actually touches.

Lemma A.1 (Pointed column-type normal form)**.** *Let $D = O^\perp \le L^N$ have L-dimension s, and choose an isomorphism $G : L^s \to D$. There are L-linear coordinate covectors $a_h \in (L^s)^\vee$ such that*

$$G(x)_h = a_h(x) \qquad (x \in L^s, 1 \le h \le N).$$

The outer code projects nontrivially onto the distinguished block j if and only if the external covectors $\{a_h : h \ne j\}$ span $(L^s)^\vee$. Conversely, every pointed list $(a_j; (a_h)_{h\ne j})$ with spanning external part realizes such an outer context, uniquely up to change of basis in L^s.

Proof. Take a_h to be the hth coordinate functional of G. The target projection of $O = D^\perp$ vanishes exactly when the target coordinate line is contained in D. Under G, this happens exactly when the common kernel of the external covectors contains a vector on which a_j is nonzero. Since G is injective, the common kernel of all a_h is zero, so this is equivalent to the external covectors failing to span. Conversely, the evaluation map defined by a pointed list with spanning external part is injective and has no nonzero target-supported image; taking its image as D gives the required outer code. Changing the basis of L^s changes all covectors by the same linear automorphism and changes nothing else. □

For rank-one targets, the exact quotient of the labelled functions observable by outer concatenation has a projective description. Informally, two rank-one inner recovery problems are indistinguishable under every compatible outer concatenation exactly when they agree on the zero-sector cost and on the cheapest response to every projective functional line an outer code can probe. We now encode those probes. Fix a line $T = \langle u \rangle \le W_P$. For $b \in L$, let $\tau_b : T \to L^*$ be defined by

$$\tau_b(au)(x) = a \operatorname{Tr}_{L/\mathbb{F}_q}(bx) \qquad (a \in \mathbb{F}_q,\ x \in L),$$

and put $z_{I,T} = \rho_T(I) + d(I^\perp)$. If $[b] = [b_1 : \cdots : b_N] \in \mathbf{P}(L^N)$ has a nonzero coordinate outside the target block j, its projective class records an outer functional-dual line. Multiplying by $s \in L^\times$

changes the chosen generator of that line but not the outer context, so the probe minimizes over this scalar choice:

$$\Theta_{I,T}([b]) = \min_{s\in L^\times} \left(\mu_{I,P,T}(\tau_{sb_j}) + \sum_{h\neq j} \lambda_{I,T}(\tau_{sb_h}) \right). \tag{41}$$

Changing u or the projective representative of $[b]$ does not change this value.

Here an *outer context for the rank-one target* T is a triple (N, j, O) with $N \geq 2$, $O \leq L^N$ an L-linear code, and nonzero projection onto the distinguished block j. The context family ranges over every such N and j, while the inner target/helper split and the identified target line T remain fixed. Thus "rank one" refers to $\dim_{\mathbb{F}_q} T = 1$, not to the dimension of O. For this numerical escape observable, two inner representations are indistinguishable at T when every compatible outer code assigns them the same equation escape cost. The theorem below identifies exactly which numerical data determine that response. Two represented inner codes are *contextually equivalent at* T when their values of $\Gamma_{j,T}$ agree in every member of this family. A compatible further concatenation adds outer layers above the same distinguished leaf. Throughout such an evaluation, the confinement region remains the entire original I-block containing the target: it is not enlarged to a newly formed composite block. Target coordinates and normalization also remain fixed. Compatibility requires that the composed outer code, viewed over L, belongs to the context family just defined.

Theorem A.2 (Minimal contextual quotient at rank one)**.** *Let $D = O^\perp \leq L^N$ under the trace identification, and suppose the projection of O onto block j is nonzero. Then*

$$\Gamma_{j,T}(O, I) = \min \left\{ z_{I,T},\ \min_{\substack{Lb\leq D\\ b\neq 0}} \Theta_{I,T}([b]) \right\}, \tag{42}$$

where Lb is the one-dimensional L-subspace spanned by b, and the inner minimum is empty when $D = 0$.

Consequently, represented inner codes over the same extension field with identified target lines are contextually equivalent at T if and only if their values $z_{I,T}$ agree and the family of truncated profiles

$$[b] \longmapsto \min\{z_{I,T}, \Theta_{I,T}([b])\}$$

agrees for every $N \geq 2$, target block j, and projective tuple with nonzero external part. Thus equality of this response family realizes the exact numerical contextual quotient at rank one: no coarser numerical summary can predict the costs in every outer context. It is enough to test the full outer code and outer codes whose functional dual has L-dimension one. The bounded higher-rank theorem below also shows that, after truncation at any fixed helper radius, only finitely many of these probes are needed. Moreover, equality of responses persists when the outer contexts are themselves formed by compatible further concatenation, with the original leaf confinement region fixed.

Proof. A nonzero map from the one-dimensional $\mathbb{F}_q$-space T into D is determined by one nonzero tuple $b \in D$. Partition $D \setminus \{0\}$ into its L-lines in the nonzero-functional sector of (9). Minimization on the line Lb is exactly (41), which proves (42).

It remains to show that every stated probe is an actual outer-code context. The full outer code $O = L^N$ has $D = 0$ and exposes $z_{I,T}$. Given a projective tuple $[b]$ with nonzero external part, take $D = Lb$ and $O = D^\perp$. Its target projection is nonzero: otherwise D would contain the target coordinate line, forcing every external coordinate of b to vanish. This context exposes $\min\{z_{I,T}, \Theta_{I,T}([b])\}$. Equality in every context therefore forces equality of the displayed response family; the first part of the theorem gives the converse.

For the closure statement, let A and C form a compatible outer context $C \circ A$. Associativity identifies $C \circ (A \circ I_i) = (C \circ A) \circ I_i$ at the physical leaf coordinates. On both sides, measure escape from the same original I_i-block, with the same target coordinates and normalization. Thus parenthesization preserves the recovery equations, their helper unions, and the requirement of a nonzero coefficient outside that region. Since $C \circ A$ is an L-linear member of the quantified context family, contextual equivalence gives equal responses for I_1 and I_2 in this context. $\square$

This theorem compares numerical costs only. Full coefficient witnesses still require the minimizing lifts described after Theorem 4.4. It does not assert congruence of escape summaries recomputed at a composite block boundary. In the binary code (a, a, a, a, b, b, b, b) with target coordinate 1, escape from $\{1, 2\}$ costs one helper, using $x_1 + x_3 = 0$. Escape from $\{1, 2, 3, 4\}$ costs three: parity requires one helper in that region and at least two outside it, attained by $x_1 + x_2 + x_5 + x_6 = 0$.

When $\dim_{\mathbb{F}_q} T = t$, the image of any map $B : T \to \mathrm{FD}(O)$ lies in an L-subspace generated by at most t images. A bounded-cost map also uses only boundedly many external labels. Together these observations give a finite small-context theorem. For $c \in \mathbb{N} \cup \{\infty\}$ write $[c]_{r+1} = \min\{c, r+1\}$, with $[\infty]_{r+1} = r + 1$.

Theorem A.3 (Rank- and radius-bounded outer tests)**.** *Let I be a represented inner code with $0 < \dim I < |E|$, let $0 \neq T \leq W_P$ have $\dim_{\mathbb{F}_q} T = t$, and let $r \geq 0$. Put $D = O^\perp \leq L^N$, where $N \geq 2$ and O projects nontrivially onto block j. For L-subspaces $D' \leq D$, set $O_{D'} = (D')^\perp \leq L^N$. Then*

$$\Gamma_{j,T}(O, I) = \min_{\substack{D' \leq D \\ \dim_L D' \leq t}} \Gamma_{j,T}(O_{D'}, I). \tag{43}$$

Moreover, for $S \subseteq [N] \setminus \{j\}$ put $D_S = D \cap L^{\{j\} \cup S}$ and $O_S = D_S^\perp \leq L^N$, where L^A denotes the vectors supported on A. Then

$$[\Gamma_{j,T}(O, I)]_{r+1} = \min_{\substack{S \subseteq [N] \setminus \{j\} \\ |S| \leq r}} [\Gamma_{j,T}(O_S, I)]_{r+1}. \tag{44}$$

Consequently, if two such inner problems have different truncated responses in some compatible outer context, a separating context exists with length at most $\max\{2, r+1\}$ and functional-dual dimension at most $\min\{t, r\}$. The target leaf and target image are kept fixed under the identifications used to compare the two problems. The same bounds cover witnesses: every nonconfined normalized coefficient-level system of cost at most r restricts to such a context and zero-extends back to the original system.

Proof. For $B : T \to D$, let $D_B = \mathrm{span}_L B(T)$. The image of an $\mathbb{F}_q$-basis of T generates D_B over L, so $\dim_L D_B \leq t$. Regarding B as a map into D_B changes none of its block labels or costs. Conversely, every map into $D' \leq D$ is a map into D. Minimizing the same cost functional in both directions proves (43); $D' = 0$ supplies the zero-functional sector. No $D' \leq D$ contains a nonzero target-supported vector, so every $O_{D'}$ retains nonzero target projection.

For the truncated identity, a map $B : T \to D$ of cost at most r has at most r nonzero external labels. If S is their set, then $B(T) \leq D_S$, and all block costs are unchanged on passing to O_S. Conversely every map into D_S is a map into D. These two inclusions prove (44); maps of larger cost all truncate to $r + 1$.

Suppose now that two truncated responses differ, and choose a minimizing map for the smaller one. If it is in the zero-functional sector, the full outer code of length two already separates the two inner problems. Otherwise let S be its nonzero external-label set and let $D_B = \mathrm{span}_L B(T)$. Then $1 \leq |S| \leq r$, $\dim_L D_B \leq t$, and the external-coordinate projection is injective on D_B, whence

$\dim_L D_B \le |S|$. Delete every coordinate outside $\{j\} \cup S$ and take the punctured D_B as the new functional dual. The chosen cost is unchanged. Since this is a subcontext of the original one, the other inner problem cannot acquire a smaller value than it had there: every map into the punctured D_B zero-extends to a map into the original D. The resulting context therefore still separates, and it has the claimed length and functional-dual dimension.

Finally, let Y be a nonconfined normalized system of cost at most r, put $B_h = \Phi_I Y_h$, and retain the target block together with the external blocks on which $Y_h \neq 0$. There are at most r of the latter. Replace the functional dual by $\operatorname{span}_L B(T)$ and delete the other blocks. The preceding dimension argument gives the same $\min\{t, r\}$ bound. Keeping activity at the coefficient level, rather than retaining only blocks with $B_h \neq 0$, preserves zero-label inner-dual lifts. Thus Y restricts to the smaller context and zero-extension recovers it exactly. □

Over the finite fields fixed here, choose one representative of every compatible pointed context within the two bounds in the theorem, and list its truncated response. This gives a finite vector $\mathcal{R}_{r,T}(I)$.

Corollary A.4 (Finite coarsest bounded contextual quotient)**.** *Two represented inner target/helper problems over the same base and extension fields, each with* $0 < \dim I < |E|$ *and with identified target normalizations on a nonzero space* T*, have equal values of* $\mathcal{R}_{r,T}$ *if and only if their truncated nonconfinement costs agree in every compatible finite outer context. Equality of this vector is therefore the coarsest numerical equivalence that determines exact responses through helper radius* r*. Equality persists under evaluation against composed outer contexts in the same family, with the original leaf confinement region, target coordinates, and normalization fixed.*

For exact, untruncated comparison, put $z_{I,T} = \rho_T(I) + d(I^\perp)$*. This is finite for every nonzero helper-recoverable* T*. Two such inner problems are contextually equivalent in all finite outer contexts if and only if their* z *values agree, say at* z*, and their exact responses agree on contexts of length at most* $\max\{2, z\}$ *and functional-dual dimension at most* $\min\{t, z-1\}$*.*

Proof. The theorem says that any unequal bounded responses have a representative among the listed contexts; the converse is immediate. Hence equality of the finite response vector is precisely observational equivalence, and any numerical summary valid in all contexts must refine it. Evaluation against a compatible composed outer context $C \circ A$ is already among these tests. Associativity and trace transitivity preserve the physical leaf coordinates and target normalization; fixing the original leaf region also preserves which equations leave it. Equality therefore persists in these evaluations, without asserting equality of newly formed macroblock escape summaries.

The full outer code of length two has response $z_{I,T}$ and separates unequal z values. When both values equal z, every exact response is at most z. Apply the theorem with $r = z - 1$: equality after truncation at z is then exact equality, and the stated finite bounds follow. □

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